

Measuring the Health-Protection Gap and Actuarially Pricing Informal-Sector Health Insurance under Nigeria's NHIA Act 2022

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Word count. About 5,500 excluding abstract, tables and references.

Abstract

Background. Nigeria's National Health Insurance Authority Act 2022 mandates health insurance and creates a Vulnerable Group Fund, but informal-sector workers remain almost entirely uncovered. No study has priced the statutory package, tested the solvency of a pool providing it, and measured the protection it would buy.

Methods. Using the General Household Survey-Panel 2023/24 (4,685 households, 24,629 individuals), we estimated catastrophic health expenditure (CHE) on a consumption aggregate calibrated to published wave-4 data, with bounds for the annualized outpatient recall. We priced the NHIA basic package from individual cost models with induced demand and simulated pool ruin under alternative take-up, contribution schedules and reinsurance, with design-based bootstrap intervals.

Results. Insurance covers 1.07% of individuals. CHE at the 10% budget-share threshold affects 14.1% of the population (6.7% to 14.1% across recall bounds; 16.7% uncalibrated) and 7.9% on capacity to pay, the only measure concentrated among the poor. The gross premium is ₦27,247 per person-year (95% interval ₦24,270 to ₦30,388) against a contribution ceiling of ₦10,022. A 20,000-life pool needs a subsidy of ₦15,339 per enrollee under random take-up and ₦51,401 under strong adverse selection at low take-up. Graded contributions halve the subsidy; reinsurance does not reduce it. Fully subsidized coverage cuts CHE to 4.4%; a flat contribution leaves it at 11.5%.

Conclusions. The statutory package costs 2.7 times what intended beneficiaries can pay. The gap is expected cost, not volatility: enrollment mechanism, contribution design and the size of the Vulnerable Group Fund decide whether coverage protects poor households.

Keywords. catastrophic health expenditure; informal sector; health insurance pricing; Tweedie model; ruin probability; universal health coverage; Nigeria

1. Introduction

Out-of-pocket payments financed 71.9% of Nigeria's current health expenditure in 2023, among the highest shares recorded anywhere (WHO Global Health Expenditure Database). The National Health Insurance Authority (NHIA) Act 2022 was meant to change this: it makes health insurance

mandatory, requires every state to run a scheme, obliges employers with five or more staff to enroll their workers, and creates a Vulnerable Group Fund to pay for those who cannot pay for themselves (Federal Republic of Nigeria 2022). Four years on, the Authority reports about 22 million enrollees, under 10% of the population, and informal employment stands at 93.0% of all employment (National Bureau of Statistics 2024). In the 2023/24 survey year analyzed here, 1.07% of individuals held health insurance.

A substantial Nigerian literature measures catastrophic health expenditure (Aregbeshola and Khan 2018; Edeh 2022; Opeloyeru and Lawanson 2023; Aniebo et al. 2025), and a commentary literature assesses the Act (Ipinnimo et al. 2022; Ahmad and Lucero-Prisno 2022). Neither tells a state scheme what the statutory package costs per enrollee, how much of that cost informal workers can carry, how large a subsidy keeps a voluntary pool solvent, or how much financial protection the package would deliver once the newly insured use more care. Those are actuarial questions, and this paper answers them on one nationally representative dataset.

The paper makes three contributions. First, it measures the protection gap under both the budget-share and the capacity-to-pay definitions, with an explicit treatment of the two measurement choices that dominate Nigerian estimates: the annualization of a four-week recall window and the construction of the consumption denominator. Second, it builds a transparent actuarial premium for the NHIA basic package from individual cost models, with design-based bootstrap intervals. Third, it embeds the cost distribution in a collective-risk model and asks what subsidy, take-up rate, contribution schedule and reinsurance arrangement a state pool needs to stay solvent, then recomputes catastrophic expenditure under each financing design. The central finding is that the gap between the actuarial price and what informal households can pay is an expected-value gap, not a volatility gap: it cannot be pooled, reinsured or diversified away, and it is multiplied by voluntary enrollment.

2. Background

2.1 *The NHIA Act 2022*

The Act replaced the National Health Insurance Scheme with an Authority and changed four things that matter here. Coverage became mandatory in law. Each state became the operative risk-bearing unit through a State Health Insurance Scheme. Employers with five or more staff were obliged to enroll their workers, a threshold we use to define formal employment. And the Vulnerable Group Fund, financed in part from the Basic Health Care Provision Fund, was created to subsidize those who cannot contribute. Ipinnimo et al. (2022) review the Act's provisions and list low government funding, workforce shortages and enforcement of the mandate as its main implementation risks; Ahmad and Lucero-Prisno (2022) call for an explicit definition of the vulnerable group and sustainable funding for the Authority. Neither attaches a number to the subsidy the Act's design requires. Several states have announced informal-sector contribution rates; Lagos State's Ilera Eko plan charges ₦15,000 a year for an individual and ₦55,000 for a family of four (July 2024). These rates are set administratively rather than actuarially, and we compare them with the modeled premium. A survey of insurance firms in Lagos, on which the present author worked, assessed social insurance as a strategy for extending social protection in Nigeria (Shogunro et al. 2023).

2.2 *Measuring catastrophic expenditure*

Two traditions coexist. The budget-share approach flags a household when health spending exceeds a fixed share of total consumption; it underlies SDG indicator 3.8.2 at the 10% and 25% thresholds and the global monitoring of Wagstaff et al. (2018) and WHO and World Bank (2023). The capacity-to-pay approach of Xu et al. (2003) compares health spending with consumption net of subsistence needs. Cylus et al. (2018) show for 14 European countries that the budget-share measure identifies richer households more often than poorer ones, while a normative-subsistence denominator concentrates catastrophic spending among the poor; Quintal (2019) makes the same point for Portugal and argues that distribution must be reported alongside incidence. Wagstaff and van Doorslaer (2003) introduced the overshoot measures used here, O'Donnell et al. (2008) codified their survey-based estimation, and Erreygers (2009) supplies the correction for binary outcomes. Shrimet et al. (2015) show that catastrophic-expenditure estimates are sensitive to the definition adopted.

For Nigeria, published incidence at the 10% threshold ranges from 16.4% for 2009/10 (Aregbeshola and Khan 2018) to 45.5% for 2023/24 (Aniebo et al. 2025), with Edeh (2022) reporting a rise across three earlier survey rounds. The last of these uses the same survey wave as this paper, and Section 6 reconciles the two.

2.3 *Informal workers, willingness to pay, and pricing*

Extending coverage to informal workers is the binding problem for universal coverage in most low- and middle-income countries. Contributions cannot be collected through payroll and enrollment is voluntary in practice whatever the statute says, so who joins is decided by the household. Onasanya (2020) argues for demography-specific communication to raise informal-sector enrollment in Nigeria. Artignan and Bellanger (2021) review 16 studies of community-based health insurance in sub-Saharan Africa and find strong evidence that it raises outpatient use, weak evidence for inpatient use, and little evidence on equity. Jofre-Bonet and Kamara (2018) elicit willingness to pay among informal workers in Sierra Leone and find a mean of about USD 3.6 per adult per month, close to 5% of average monthly business income; they do not cost the package such a premium would fund. The 5% share is the same figure used as the affordability ceiling here.

Aggregate annual health cost per person has a point mass at zero and a long right tail, the structure the Tweedie compound Poisson-gamma family was built for (Smyth and Jørgensen 2002; Dunn and Smyth 2005). Klugman, Panjer and Willmot (2019) give the collective-risk and premium-principle framework, and Wüthrich (2015) links classical ruin probability to solvency capital. The demand response to insurance is the parameter that most affects a price: the RAND Health Insurance Experiment gives an arc elasticity near -0.2 (Manning et al. 1987), and Boes and Gerfin (2016) estimate about -0.14 from a Swiss policy change together with a rise in the probability of any cost.

3. Data

Source and sample. The Nigeria General Household Survey-Panel (GHS-Panel) is collected by the National Bureau of Statistics with the World Bank LSMS-ISA program. Wave 5 (2023/24) is the analysis file; wave 4 (2018/19), which carries a published consumption aggregate, is used for calibration and comparison. After dropping households without a cross-sectional weight, the file holds 4,685 households and 24,629 individuals in 511 enumeration areas across six strata; the weight grosses to 40.6 million households and 211 million people. Household characteristics are weighted by

the household weight; catastrophic-expenditure rates weight households by household size, so they are the share of people living in an affected household, following the SDG 3.8.2 convention. Household-weighted rates are reported alongside.

Out-of-pocket spending. The health module asks every member whether they sought care in the past four weeks and what was paid for the consultation (explicitly excluding drugs), prescription drugs and non-prescription drugs, and separately about hospitalization and its cost over twelve months. The drug questions exclude drugs related to admissions and the hospitalization question includes them, so the two components do not overlap. Drugs are 86.25% of measured outpatient spending. Transport is excluded, following WHO practice, and tested as a variant. The non-food consumption module also asks a single twelve-month question about health spending; only 17% of households report any, implying a spending share of 0.3% of consumption against 4.9% in the published wave-4 aggregate, so it is not used.

Annualization. Outpatient spending observed in a four-week window is scaled by 13; inpatient spending is already annual. Scaling the window assumes the other twelve windows repeat the observed one, which maximizes the dispersion of annual spending across households and therefore the share crossing any threshold. The published wave-4 aggregate scales its own one-month health item by about twelve, so the convention is shared with the official statistics. Section 5.6 bounds its effect.

Consumption. Wave 5 publishes no consumption aggregate. One is rebuilt from the food module (seven-day recall, own production and gifts valued at local median unit values), three non-food modules and the education module, with health spending entering once from the health module. The same code run on wave 4 recovers 84% of the published aggregate, ranks households with a Spearman correlation of 0.80, and omits imputed rent. Holding out-of-pocket spending fixed, that shortfall raises CHE at the 10% threshold on wave 4 from 14.1% with the official aggregate to 17.6% with the rebuilt one. The main results therefore use a calibrated aggregate: each rebuilt component is scaled by the ratio of its official to its rebuilt wave-4 mean (1.158 for food, 0.892 for non-food) and imputed rent is added at the official rent-to-non-rent ratio (0.069). Imputed rent is 6.4% of calibrated wave-5 consumption. Applied to wave 4 itself the calibrated aggregate recovers 96% of the official level and gives CHE of 15.2% against 14.1% (Table A1). Uncalibrated results are reported throughout so the effect of the calibration is visible.

Coverage and sector. Section 5A asks whether any household member holds insurance, of what type, and which members are covered; the health type was identified by the fact that all ten households reporting a health-insurance premium in the consumption module report type 1. A worker is formal when the employer is a public body, state-owned enterprise, NGO or international organization, or a private firm with five or more staff; a household is informal if it contains no formal worker. On this rule 81.2% of households are informal (80.0% of the population). Three alternative rules are tested.

Prices and poverty line. All naira are constant August-2023 prices. The 2019 national poverty line of ₦137,430 per person per year (National Bureau of Statistics 2020) is carried forward with the composite CPI (factor 2.31) to ₦317,463.

4. Methods

4.1 Catastrophic expenditure and inequality

Under the budget-share definition a household is catastrophic if OOP/C exceeds 10%, 25% or 40% of total consumption C . Under the capacity-to-pay definition (Xu et al. 2003) household size is equivalized as $size^{0.56}$, subsistence spending per equivalent adult is the weighted mean food spending of households whose food share lies between the 45th and 55th percentiles, capacity to pay is consumption net of subsistence (or net of actual food spending where that is smaller), and the threshold is 40% of it. Intensity is the mean overshoot among affected households (Wagstaff and van Doorslaer 2003). Impoverishment compares poverty headcounts gross and net of out-of-pocket payments.

Inequality is summarized by the concentration index, $CI = 2 \text{ cov}(y, r)/\text{mean}(y)$, where r is the weighted fractional rank in per-capita consumption, with the Erreygers (2009) correction for binary outcomes and design-based standard errors from linearizing each index as a function of totals. All standard errors are Taylor-linearized for a stratified single-stage cluster design with enumeration areas as primary sampling units (Lumley 2004); domain estimates zero the weights outside the domain rather than dropping rows. Changes between scenarios are tested on the household-level difference, since the same households appear under both regimes.

4.2 Cost models

The unit is the individual-year and the models are survey-weighted with standard errors clustered on the enumeration area. Outpatient contact within the four-week window is modeled as a Poisson count with an offset of $\log(4/52)$, so fitted values are annual rates; inpatient episodes are modeled over twelve months without an offset. Cost per episode is modeled with gamma and lognormal GLMs. Annual cost per person is modeled with a Tweedie compound Poisson-gamma GLM with a log link, the variance power chosen by profile likelihood over 1.20 to 1.85; its fitted values are the predicted cost that drives the adverse-selection scenarios. Covariates are age band, sex, zone, urban residence, consumption quintile, severe functional difficulty from the Washington Group short set, and household size. Diagnostics are randomized quantile residuals (Dunn and Smyth 1996) and an observed-against-predicted lift table.

4.3 Benefit mapping and premium

The NHIA tariff schedule is not published in machine-readable form, so the package is defined by parameters, each a robustness lever. Traditional and spiritual care is excluded. Of the remainder, 85% of outpatient and 90% of inpatient spending is assumed to fall inside the package. Covered outpatient spending is split into drugs, which carry the NHIA's 10% co-payment, and services, which are free at the point of use. Induced demand uses the arc (midpoint) elasticity: with $e = -0.20$ and the patient's price falling from 1.0 to zero, use rises by a factor $(1 + x)/(1 - x)$ with $x = e(p_1 - p_0)/(p_0 + p_1)$, which is 1.50 for services and 1.39 for drugs. The insurer's expected cost per enrollee is the pure premium. A risk margin is added on the standard-deviation principle ($0.10 \sigma/\sqrt{n}$) or, alternatively, as a 95% CVaR margin on the simulated pool mean; an administrative load of 15% and an adverse-selection load of 10% then give the gross premium. Affordability is judged against 5% of per-capita consumption; the contribution ceiling for the target population is 5% of the mean per-capita consumption of informal-sector members of the bottom three quintiles.

4.4 Pool solvency and financing designs

A discrete-time collective-risk model tracks the surplus of a pool of n enrollees, $U_t = U_{(t-1)} + n(C + s)(1 - e) - S_t$, where C is the mean member contribution, s the subsidy per enrollee, e the expense ratio (10%), and S_t aggregate claims in year t , drawn by resampling annual insurer costs from the informal-sector population with survey weights (Klugman et al. 2019). Ruin is a negative surplus in any year of a one- or three-year horizon. The minimum subsidy holding the ruin probability below 5% or 1% is found by bisection on a fixed set of 10,000 simulated claim paths, so every financing scenario faces the same simulated experience. The empirical cost distribution is collapsed to its distinct values and each pool-year is drawn as one multinomial, which is exact and is checked against direct resampling (Table A12).

Adverse selection tilts enrollment odds by a factor of 1.5 or 2.0 per standard deviation of log predicted cost. Because a tilt is only possible when some people stay out, a take-up grid sets the enrollment probability to $\min(1, k \times \text{tilt})$ with k chosen so mean take-up equals 10%, 25%, 50%, 75% or 100%; at 100% take-up no selection is possible, which is the mandatory case. Three contribution schedules are compared: flat (the bottom-three-quintile ceiling for everyone), graded (each quintile's own ceiling) and exempt (Q1 and Q2 pay nothing, Q3 to Q5 graded). Per-life excess-of-loss reinsurance at retentions of ₦250,000, ₦500,000 and ₦1 million is priced at 1.3 times the expected ceded cost. A permanent 25% rise in claims from year 2 is the stress scenario.

4.5 Coverage counterfactual

Covered individuals pay the drug co-payment plus anything outside the package on the higher utilization induced demand implies; uncovered individuals keep their observed spending. Non-health consumption is held fixed and any contribution is a new call on the same budget. Catastrophic expenditure is recomputed on two bases: out-of-pocket payments alone (the SDG convention) and out-of-pocket payments plus the contribution. The exercise is an accounting simulation, not a causal estimate.

4.6 Recall bounds and bootstrap

The annualization is bounded by two extremes. The upper case scales each person's observed window to a year (full within-year persistence). The lower case draws each person's annual outpatient cost as a sum of N independent four-week windows, $N \sim \text{Poisson}(13 \times \hat{p})$, where $\hat{p}$ is the fitted contact probability, with episode costs resampled from observed episodes in the person's quintile; 20 replicates are averaged. Both are carried through CHE, the premium and the subsidy, with the $\times 12$ and $\times 10$ multipliers alongside. A Rao-Wu rescaled bootstrap over enumeration areas within strata (200 replicates) gives percentile intervals for the premium, the expected claim and the minimum subsidy, quantities for which no linearized standard error exists.

5. Results

5.1 Sample, coverage and catastrophic expenditure

Mean household size is 5.2, mean annual consumption on the calibrated aggregate ₦1.99 million, and out-of-pocket health payments average 4.9% of consumption (Table 1). Informal-sector households are poorer (per-capita consumption ₦524,747 against ₦609,625, $p = 0.008$), more rural (34.3% against 60.1% urban) and more often female-headed (22.7% against 11.4%). Health insurance reaches 1.98%

of households and 1.07% of individuals; coverage is 8.03% among formal-sector households against 0.57% among informal-sector households, and 0.20% of households paid a premium themselves (Table A2). The premise that the informal sector is the uncovered sector is observed, not assumed.

Catastrophic expenditure affects 14.1% of the population at the 10% threshold (95% CI 12.5 to 15.8), 3.4% at 25% and 0.8% at 40%; 7.9% (6.7 to 9.1) exceed 40% of capacity to pay (Table 2, Figure 1). The household-weighted rate at the 10% threshold is 14.8%. On the uncalibrated aggregate the same figures are 16.7% and 11.3%; the calibration lowers both because it raises the denominator. Affected households do not cross the line narrowly: the mean overshoot is 10.2 points at the 10% threshold and 15.1 points on capacity to pay. Out-of-pocket payments raise the poverty headcount from 57.2% to 60.3%, an impoverishing effect of 3.1 points that is stable across poverty lines from half to one and a half times the national line while the headcount itself is not (Table A3).

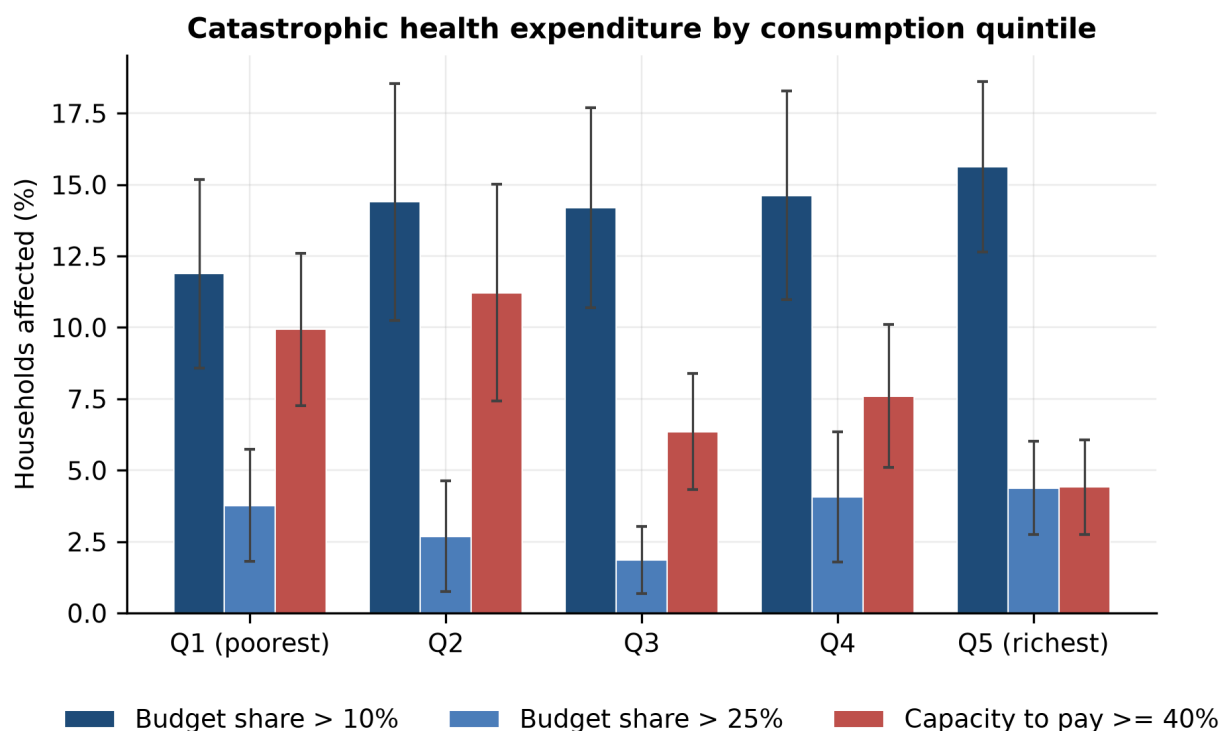


Figure 1. Catastrophic health expenditure by consumption quintile at the 10% and 25% budget-share thresholds and the 40% capacity-to-pay threshold. Bars are survey-weighted incidence; whiskers are 95% design-based confidence intervals.

5.2 The two definitions disagree about who is affected

The concentration index for budget-share CHE at 10% is +0.040 (95% CI −0.018 to +0.098), not distinguishable from zero; for capacity-to-pay CHE it is −0.150 (−0.223 to −0.077), pro-poor (Table 3, Figure 2). The same pattern appears by sector: on the budget-share measure informal and formal households are equally affected (14.4% against 13.3%, $p = 0.59$), while on capacity to pay informal households are affected at 8.7% against 4.8% ($p = 0.005$). Most of that gap is composition. In a population-weighted linear probability model the informal coefficient on capacity-to-pay CHE falls from 3.8 points unadjusted to 3.1 points ($p = 0.03$) with quintile controls and to 2.4 points ($p = 0.11$) with quintile, zone, residence and household-size controls; on budget share it is 1.3 points ($p = 0.49$) with the same controls (Table 3). Informal households are more exposed because they are poorer and

more rural, not detectably because of informality itself. The mechanism is arithmetic. Richer households consume more and buy more care, so the ratio to total consumption does not sort by hardship; net of subsistence, the same naira registers as a much larger shock for a poor household. A linear decomposition (Table A4) attributes most of the budget-share index to hospitalization and household size, but the index is too close to zero for shares to be meaningful, and for the capacity-to-pay index the residual dominates, so the decomposition is reported for completeness only.

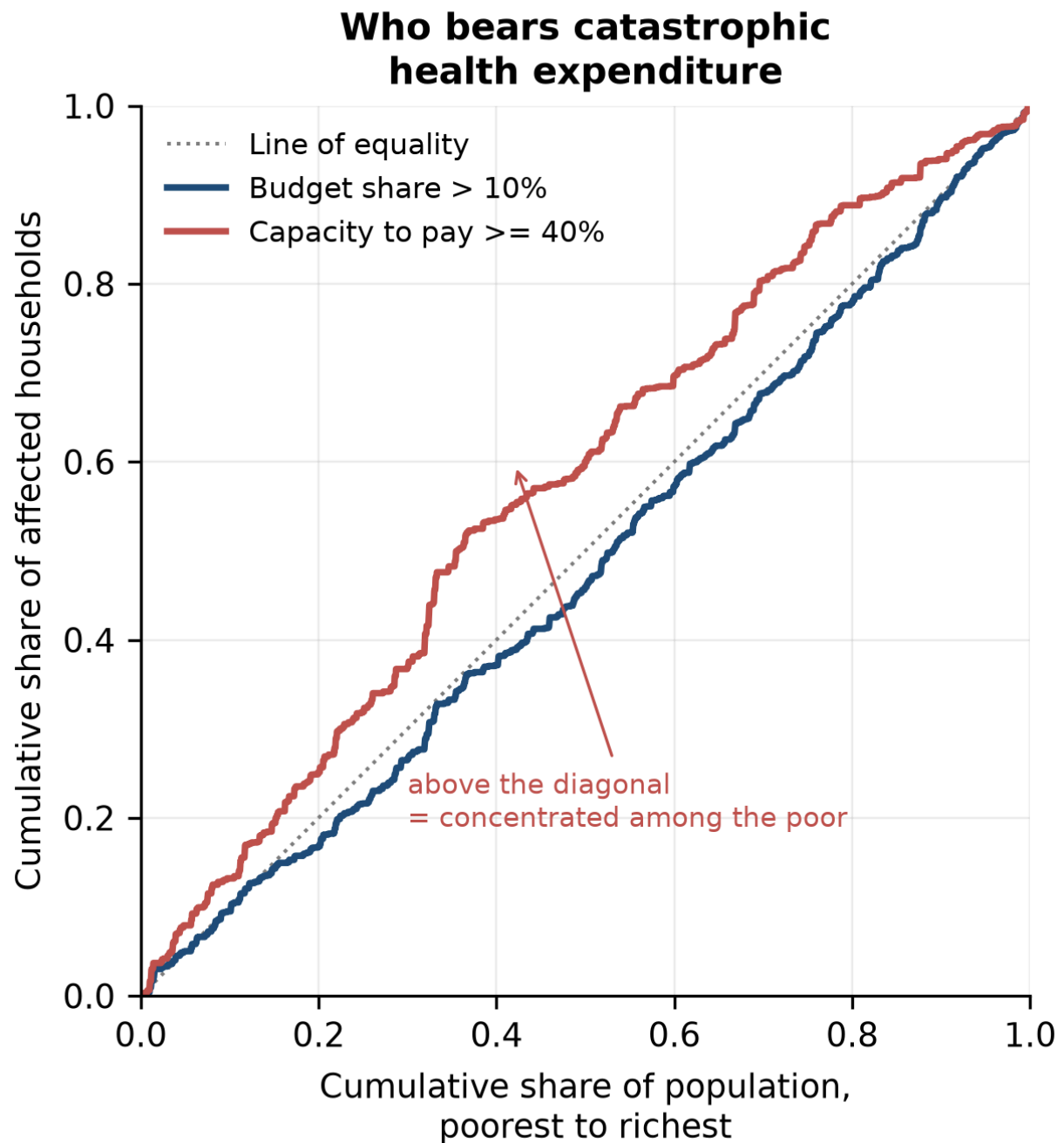


Figure 2. Concentration curves for the two definitions. A curve above the diagonal indicates a burden concentrated among the poor.

5.3 Cost models

Individuals average 2.75 outpatient contacts and 0.030 inpatient episodes a year, at mean costs of ₦6,977 per outpatient episode and ₦43,985 per inpatient episode, giving a mean annual cost of ₦19,675 with 78% of individuals recording no cost. The contact model is underdispersed (Pearson dispersion 0.78), as expected when a binary indicator is modeled as a count, so the negative binomial adds nothing. The lognormal fits episode cost better than the gamma (AIC 95,299 against 98,131). The profile likelihood selects a Tweedie variance power of $p = 1.65$ (Tables A6 and A7). Calibration by decile of prediction is close, with observed-to-predicted ratios between 0.81 and 1.13 and a top-decile mean of ₦83,195 against a bottom-decile mean of ₦3,104. Randomized quantile residuals have mean 0.07, standard deviation 1.00 and skew -0.11 ; the normality test rejects at $n = 24,629$, as it will for any parametric model. Risk relativities are large (Table A5): people aged 60 or over cost 3.31 times the community mean and those with a severe functional difficulty 6.44 times; cost rises from 0.26 of the mean in the poorest quintile to 2.57 in the richest, a demand effect that also means observed spending understates the health needs of the poor.

5.4 The premium and what people can pay

Observed cost of ₦19,675 per person-year falls to ₦16,487 once traditional care and services outside the package are removed; induced demand adds ₦6,808 and the drug co-payment returns ₦1,837, leaving a pure premium of ₦21,459 (Table 4, Figure 3). Loadings take a 20,000-life pool to a gross premium of ₦27,247 on the standard-deviation margin and ₦29,437 on the CVaR margin. The risk margin is ₦81 and falls to ₦36 at 100,000 lives; administration and the selection load, not claims volatility, separate the gross from the pure premium. The bootstrap interval for the gross premium is ₦24,270 to ₦30,388 (Table 8).

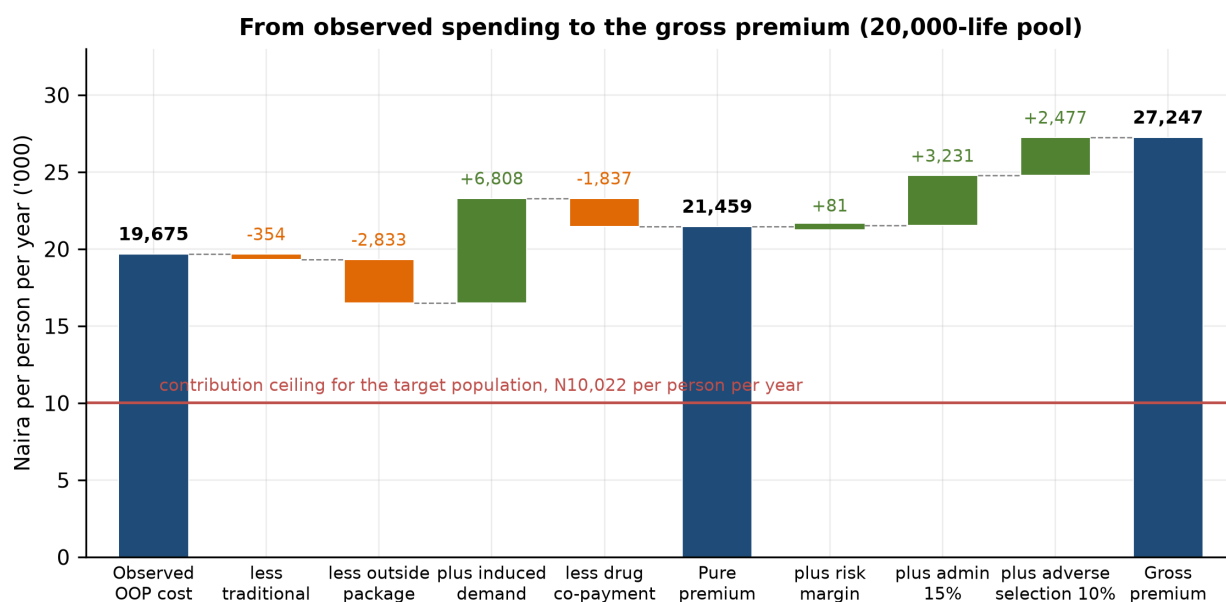


Figure 3. From observed spending to the gross premium for a 20,000-life pool. The horizontal line is the contribution ceiling for informal-sector members of the bottom three quintiles.

Measured against per-capita consumption the gross premium is 22.8% in the poorest quintile, 13.4% in Q2, 9.7% in Q3, 6.6% in Q4 and 3.0% in Q5, above the 5% ceiling in four quintiles out of five (Table 4). Billed per household it is ₦222,993 a year for an average household in Q1, which has 8.2 members. The contribution ceiling for informal-sector members of the bottom three quintiles is ₦10,022 per person-year, 37% of the gross premium. The modeled premium is 1.8 times Lagos State's individual

Ilera Eko rate and half its family-of-four rate, a comparison that is a plausibility check rather than a like-for-like test because the packages differ and the published rates are nominal 2024 naira.

5.5 Solvency, take-up and financing design

At the flat contribution of ₦10,022, a 20,000-life pool with no opening capital needs a subsidy of ₦15,339 per enrollee to hold its three-year ruin probability below 5% under random take-up, ₦16,019 for a 1% target, and ₦51,401 under strong adverse selection (Table 5, Figure 4). Most of the subsidy is expected value: the expected claim of ₦21,342 divided by $(1 - e)$ less the contribution gives ₦13,691, so process risk adds about ₦1,600 at this scale. Moving from a 5% to a 1% ruin target costs ₦680 per enrollee; moving from 5,000 to 100,000 lives lowers the subsidy under random take-up by 16%; moving from random to strong adverse selection raises it by a factor of 3.4. Opening capital equal to a quarter of a year's income at the gross premium lowers the random-take-up subsidy to ₦11,161, and a permanent 25% rise in claims from year 2 raises it to ₦18,723 (Table A8).

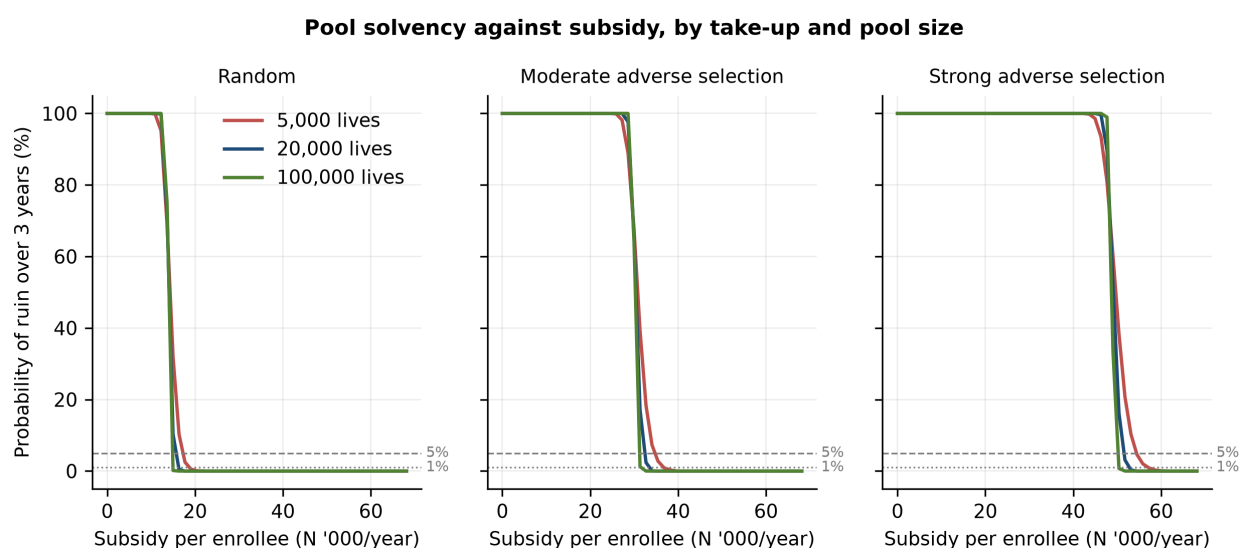


Figure 4. Probability of ruin over three years against subsidy per enrollee, by pool size and take-up pattern.

The take-up grid shows why enrollment mechanism matters more than pool size (Table 5, Figure 5). With a strong tilt the selection loading on expected claims is 147% at 10% take-up, 60% at 50%, 24% at 75% and zero at 100%; the corresponding subsidy falls from ₦51,486 to ₦21,351 at 75% take-up and ₦15,358 at full take-up. The moderate tilt converges to the same path above 50% take-up, because at high take-up almost everyone is enrolled whatever their predicted cost. The adverse-selection scenarios in the upper panel of Table 5 are therefore low-take-up scenarios, and the mandate in the Act, if enforced through group enrollment, is worth more to solvency than any plausible increase in pool size.

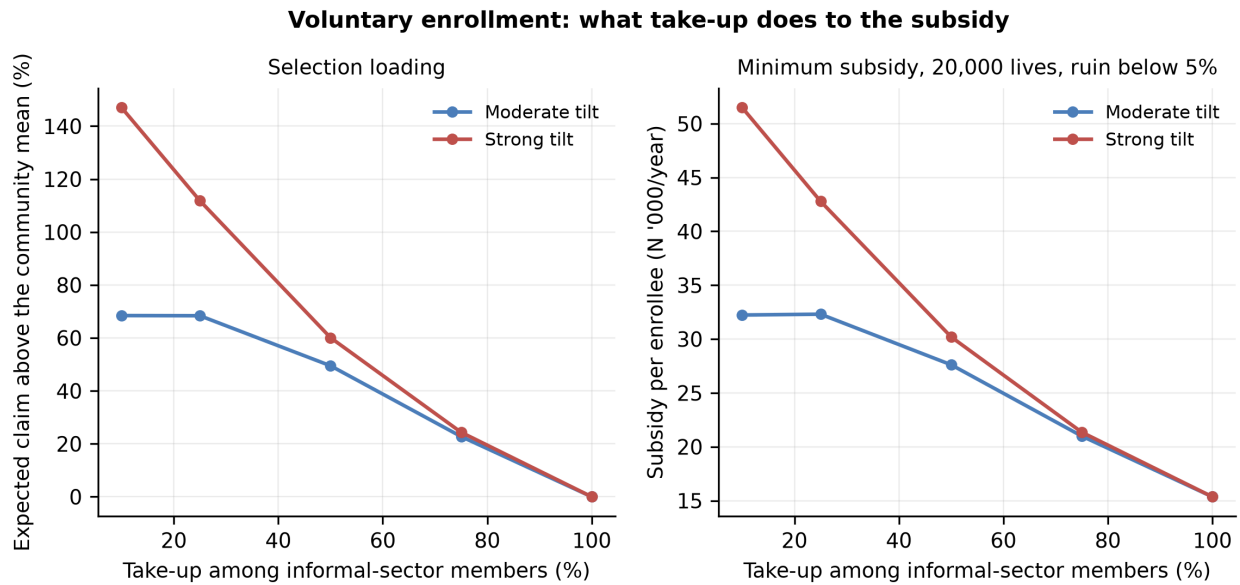


Figure 5. Voluntary enrollment: selection loading on the expected claim, and the subsidy needed for a 20,000-life pool, against the take-up rate among informal-sector members.

Contribution design changes the subsidy by more than reinsurance does (Table 6). A graded schedule in which each quintile pays its own 5% ceiling raises the mean contribution to ₦17,986 and lowers the subsidy under random take-up to ₦7,375; exempting the bottom two quintiles and grading the rest gives ₦14,529 and ₦10,831. Under strong selection the graded schedules collect more because the sick are disproportionately in the richer quintiles, and the subsidy falls from ₦51,401 to about ₦35,000 to ₦37,000. Per-life excess-of-loss reinsurance at a ₦250,000 retention cuts the standard deviation of pool claims per enrollee from ₦850 to ₦315, but the loaded reinsurance premium of ₦8,260 exceeds the process-risk saving and the required subsidy rises to ₦16,399; at higher retentions the effect is smaller in both directions. Reinsurance protects a pool against volatility, and volatility is not this pool's problem.

5.6 How much protection coverage buys

Measured on out-of-pocket payments alone, full coverage of the informal sector cuts CHE at the 10% threshold from 14.1% to 4.2% and capacity-to-pay CHE from 7.9% to 1.5% (Table 7). Measured on what households actually pay, the same coverage financed by the flat contribution cuts CHE only to 11.5% (a reduction of 2.6 points, $p < 0.001$), and capacity-to-pay CHE is unchanged at 8.4% ($p = 0.47$): the contribution removes from poor households roughly what the coverage gives them, every year rather than only in years of illness. By quintile, the flat contribution raises CHE in the poorest quintile from 11.9% to 31.0% while lowering it elsewhere (Figure 6, Tables A9 and A10). Fully subsidized coverage cuts CHE to 4.4% (a reduction of 9.7 points) and capacity-to-pay CHE to 1.7%. Exempting the bottom two quintiles and grading the rest reaches 7.6% and 2.5%, and subsidizing only the vulnerable group reaches 10.4% and 4.5%. On the out-of-pocket-only basis the contributory scheme scores marginally better than the fully subsidized one (4.2% against 4.4%) because the contribution enlarges the consumption denominator; that is a property of the SDG indicator, and it is why both bases are reported.

Catastrophic spending under coverage scenarios, out-of-pocket plus contribution

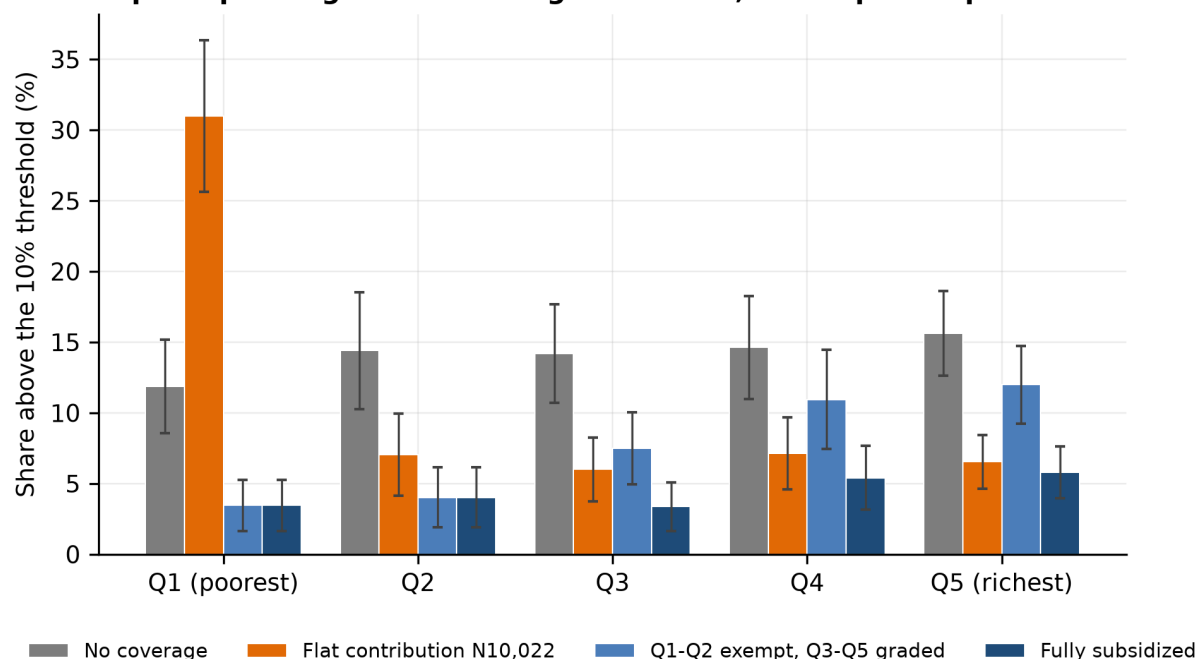


Figure 6. Catastrophic spending at the 10% threshold under coverage scenarios, by consumption quintile, on out-of-pocket payments plus the member contribution.

5.7 Bounds, intervals and robustness

The recall treatment is the largest single uncertainty (Table 8). With independent four-week windows in place of full persistence, CHE at the 10% threshold falls from 14.1% to 6.7% and capacity-to-pay CHE from 7.9% to 3.3%, because a household's annual spending is then the sum of many small pieces rather than thirteen copies of one. The premium barely moves (₦27,612 against ₦27,247), since it is a mean, and the random-take-up subsidy falls by about ₦1,500 through lower process risk and a slightly lower expected claim. Multipliers of 12 and 10 lower CHE to 13.3% and 11.2% and the gross premium to ₦25,324 and ₦21,478. The truth lies between the persistence bounds; chronic conditions make some persistence certain.

The bootstrap gives a standard error of 0.9 points on CHE at the 10% threshold, matching the linearized 0.8, and intervals of ₦24,270 to ₦30,388 for the gross premium, ₦12,050 to ₦18,948 for the random-take-up subsidy and ₦41,076 to ₦62,337 under strong selection. The 95% interval for the random-take-up subsidy is about a fifth as wide as the gap between random take-up and strong selection.

Across the 26 rows of the robustness grid (Table A11), equivalence-scale exponents from 0.5 to 1.0 move capacity-to-pay CHE by under 0.2 points; alternative informality rules change the subsidy by under 5%; fixing the Tweedie power at 1.4, 1.5 or 1.7 changes nothing material; including transport raises CHE to 15.3% and the premium by 7%; trimming the top 1% of costs lowers the premium by 20% and the subsidy by 40%, which is why the tail is kept. Benefit-design parameters behave monotonically: the elasticity range -0.10 to -0.35 spans gross premiums of ₦22,874 to ₦35,858. Wave 4 on its published aggregate gives 14.1% at the 10% threshold and 6.7% on capacity to pay (Table A12), against 14.1% and 7.9% for wave 5 on the calibrated aggregate. The waves differ in season (the wave-4 health module was fielded after harvest) and in the annualizer, so the comparison supports

no trend in either direction; the rise that the uncalibrated wave-5 figures suggest comes from the denominator.

6. Discussion

What the numbers say to a state scheme. The pure premium for the basic package is about ₦21,500 per person-year in 2023/24 prices and the gross premium about ₦27,200, with a sampling interval of roughly ₦24,000 to ₦30,000. The most the target population can contribute is about ₦10,000. The difference is the subsidy, ₦15,300 per enrollee under random take-up and up to ₦51,400 under strong selection, or ₦1.5 to ₦5.1 billion a year per 100,000 enrollees. Nine tenths of that subsidy is the expected-value gap between the price of the package and the contribution, which no pooling, reinsurance or capital arrangement can remove. The operative policy question is the size and reliability of the Vulnerable Group Fund, not the contribution rate.

Design. Three design results follow. First, selection dominates scale: enrolling whole groups (cooperatives, market associations, transport unions, which the Act's recognition of Mutual Health Associations permits) at 75% take-up or better needs less than half the subsidy of individual voluntary enrollment at 10% take-up under a strong tilt. Second, a flat contribution undoes most of the protection it finances. Capacity-to-pay CHE does not fall when coverage is paid for by a uniform ₦10,022 contribution, and CHE in the poorest quintile rises from 12% to 31%. Contributions should be graded and the bottom two quintiles exempt; under random take-up that design lowers the subsidy by 29% relative to the flat schedule (a fully graded schedule lowers it by 52%) and delivers about two thirds of the reduction in CHE that full subsidy delivers. Third, reinsurance is the wrong instrument for this pool: its premium exceeds the volatility it removes.

Measurement, and the Aniebo et al. estimate. Our 14.1% at the 10% threshold sits with Aregbeshola and Khan's 16.4% for 2009/10 and our own 14.1% for wave 4, and far from the 45.5% that Aniebo et al. (2025) report on the same 2023/24 survey, with a capacity-to-pay incidence of 43.1% and a pro-poor budget-share index of -0.152. Because both studies use the same households, the difference is a measurement difference, not a change in the population. Their abstract does not give enough detail on annualization or the consumption denominator for us to reproduce the 45.5%, so we cannot say which choice produces it. We can say how far the plausible choices move the estimate: within our own pipeline the annualization moves CHE between 6.7% and 14.1%, the denominator moves it between 14.1% and 16.7%, and the choice of instrument moves it between 0.1% and 16.7%. The level of catastrophic expenditure in Nigeria is dominated by measurement choices, and any subsidy calculation built on a single level inherits that. What is stable across our specifications is the ordering: the capacity-to-pay measure is pro-poor, and the budget-share measure shows no significant gradient in our data. Aniebo et al. find both measures pro-poor, so the budget-share gradient is itself sensitive to measurement choices, and monitoring should report both definitions rather than rely on either alone.

Limitations. The counterfactual holds total consumption fixed, applies one elasticity from a US experiment, and assumes the package is delivered as specified; Boes and Gerfin's smaller elasticity would lower the premium, while their extensive-margin effect would raise it. The package parameters are assumptions until the NHIA publishes a tariff. Observed coverage (107 households) is too thin to identify the effect of coverage on spending, so coverage is assigned rather than estimated. The consumption aggregate is calibrated to wave 4 in the mean, so wave-5 levels, and the poverty

headcount in particular, remain approximate. The selection scenarios are modeled tilts, not observed take-up. Households that forgo care appear as zero spenders, so every measure understates unmet need.

7. Conclusion

Catastrophic health expenditure affects between 7% and 14% of Nigerians at the 10% budget-share threshold depending on how a four-week recall window is treated, and between 3% and 8% on capacity to pay; only the capacity-to-pay measure locates the burden among poor households, and the higher rate among informal-sector households reflects their lower consumption. The NHIA basic package costs about ₦21,500 per person-year before loadings and ₦27,200 after, against a contribution ceiling of ₦10,000 for the households the Act intends to protect. Closing the gap is public expenditure, not insurance sold to informal workers: about ₦15,000 per enrollee under group enrollment and three times that under individual voluntary enrollment. Grading contributions and exempting the poorest two quintiles halves the subsidy and preserves most of the protection; a flat contribution preserves little of it.

Declarations

Data availability. The GHS-Panel microdata are publicly available from the World Bank Microdata Library subject to registration and are not redistributed. All analysis code is available at <https://github.com/tosin-babs/nigeria-health-protection-gap> and will be archived with a Zenodo DOI on submission. An interactive calculator that reruns the pricing and solvency model in the browser is available at <https://nigeria-health-protection-gap.vercel.app>.

Code availability. Complete, seeded reproduction code in Python at the repository above; the README gives the script order and environment. The pipeline runs end to end from the raw survey files in about fifteen minutes.

Competing interests. None declared.

Ethics. The analysis uses de-identified secondary survey data and did not require ethical approval.

AI-assistance disclosure. Generative AI (Claude, Anthropic) was used to assist with code development, code review and language editing. The author designed the study, specified all models and parameters, verified and interpreted all results, and takes full responsibility for the content. AI systems are not authors and are not listed as such.

CRedit statement. Oluwatosin Dorcas Babalola: conceptualization, methodology, software, formal analysis, data curation, writing (original draft, review and editing).

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Tables

Generated from `output/tables/.csv` by `python/make_tables.py`. All estimates are survey-weighted with Taylor-linearized standard errors for a stratified single-stage cluster design. Naira are in constant August 2023 prices. Consumption is the calibrated aggregate unless a row says otherwise.*

Table 1. Sample characteristics, weighted by household weight.

p is a design-based Wald test of the informal-formal difference.

Characteristic	Overall	SE	Informal	Formal	p
Household size	5.21	0.10	5.13	5.53	0.046
Head's age (years)	52.46	0.35	52.95	50.32	0.000
Female-headed (%)	20.60	1.00	22.73	11.43	0.000
Urban (%)	39.14	2.72	34.29	60.08	0.000
Children under 5 (n)	0.37	0.02	0.37	0.37	0.979
Members aged 60+ (n)	0.42	0.01	0.44	0.33	0.002
Any member with severe functional difficulty (%)	9.81	0.61	10.53	6.72	0.005

Characteristic	Overall	SE	Informal	Formal	p
Total annual consumption (N)	1,991,166.15	44,759.72	1,846,388.41	2,615,920.99	0.000
Consumption per capita (N)	540,716.09	15,639.67	524,747.42	609,625.22	0.008
Food share of consumption (%)	69.30	0.47	70.60	63.66	0.000
Annual out-of-pocket health spending (N)	102,443.47	6,140.09	100,606.09	110,372.25	0.414
OOP as share of consumption (%)	4.86	0.19	5.00	4.23	0.050
Any member sought care in past 4 weeks (%)	62.41	1.14	61.77	65.18	0.230
Any member hospitalized in past 12 months (%)	11.91	0.70	11.66	12.97	0.469
Any member holds health insurance (%)	1.98	0.34	0.57	8.03	0.000
Paid a health-insurance premium in past 12 months (%)	0.20	0.10	0.03	0.94	0.064

Table 2. Catastrophic health expenditure: incidence and intensity.

Population-weighted rates are the share of people living in an affected household; the household-weighted row is the share of households. Uncalibrated rows use the rebuilt consumption aggregate without the wave-4 calibration. Zone and residence breakdowns are in Table A2.

Measure	Dimension	Group	Incidence %	95% low	95% high	Mean positive overshoot pp	n
Budget share > 10%	Overall	All households	14.1	12.5	15.8	10.2	4,685
Budget share > 10%	Overall (household-weighted)	All households	14.8	13.4	16.3	11.0	4,685

Measure	Dimension	Group	Incidence %	95% low	95% high	Mean positive overshoot pp	n
Budget share > 10%	Overall, uncalibrated aggregate	All households	16.7	14.9	18.4		4,685
Budget share > 10%	Sector	Formal	13.3	9.9	16.7	9.1	809
Budget share > 10%	Sector	Informal	14.4	12.6	16.1	10.5	3,876
Budget share > 10%	Consumption quintile	Q1 (poorest)	11.9	8.6	15.2	11.9	701
Budget share > 10%	Consumption quintile	Q2	14.4	10.3	18.5	9.2	699
Budget share > 10%	Consumption quintile	Q3	14.2	10.7	17.7	7.1	808
Budget share > 10%	Consumption quintile	Q4	14.6	11.0	18.3	10.5	1,041
Budget share > 10%	Consumption quintile	Q5 (richest)	15.6	12.6	18.6	12.4	1,436
Budget share > 25%	Overall	All households	3.4	2.5	4.2	10.4	4,685
Budget share > 25%	Overall (household-weighted)	All households	3.7	3.0	4.5	12.5	4,685
Budget share > 25%	Overall, uncalibrated aggregate	All households	4.6	3.6	5.7		4,685
Budget share > 25%	Sector	Formal	2.5	0.7	4.3	7.8	809
Budget share > 25%	Sector	Informal	3.6	2.6	4.5	10.8	3,876
Budget share > 25%	Consumption quintile	Q1 (poorest)	3.8	1.8	5.7	10.4	701
Budget share > 25%	Consumption quintile	Q2	2.7	0.8	4.6	10.9	699

Measure	Dimension	Group	Incidence %	95% low	95% high	Mean positive overshoot pp	n
Budget share > 25%	Consumption quintile	Q3	1.9	0.7	3.0	4.3	808
Budget share > 25%	Consumption quintile	Q4	4.1	1.8	6.3	8.0	1,041
Budget share > 25%	Consumption quintile	Q5 (richest)	4.4	2.7	6.0	14.8	1,436
Budget share > 40%	Overall	All households	0.8	0.5	1.1	10.2	4,685
Budget share > 40%	Overall (household-weighted)	All households	1.2	0.8	1.6	11.9	4,685
Budget share > 40%	Sector	Formal	0.3	-0.1	0.7	4.9	809
Budget share > 40%	Sector	Informal	0.9	0.6	1.3	10.6	3,876
Budget share > 40%	Consumption quintile	Q1 (poorest)	1.0	0.2	1.9	5.8	701
Budget share > 40%	Consumption quintile	Q2	0.8	-0.2	1.8	8.9	699
Budget share > 40%	Consumption quintile	Q3	0.0	-0.0	0.1	2.6	808
Budget share > 40%	Consumption quintile	Q4	0.5	0.0	1.0	8.3	1,041
Budget share > 40%	Consumption quintile	Q5 (richest)	1.7	0.8	2.5	14.4	1,436
Capacity to pay >= 40%	Overall	All households	7.9	6.7	9.1	15.1	4,685
Capacity to pay >= 40%	Overall (household-weighted)	All households	7.8	6.8	8.8	16.3	4,685
Capacity to pay >= 40%	Overall, uncalibrated aggregate	All households	11.3	9.8	12.8		4,685

Measure	Dimension	Group	Incidence %	95% low	95% high	Mean positive overshoot pp	n
Capacity to pay >= 40%	Sector	Formal	4.8	2.6	7.1	13.2	809
Capacity to pay >= 40%	Sector	Informal	8.7	7.2	10.1	15.3	3,876
Capacity to pay >= 40%	Consumption quintile	Q1 (poorest)	9.9	7.3	12.6	15.6	701
Capacity to pay >= 40%	Consumption quintile	Q2	11.2	7.4	15.0	15.0	699
Capacity to pay >= 40%	Consumption quintile	Q3	6.4	4.3	8.4	12.5	808
Capacity to pay >= 40%	Consumption quintile	Q4	7.6	5.1	10.1	13.7	1,041
Capacity to pay >= 40%	Consumption quintile	Q5 (richest)	4.4	2.8	6.1	20.1	1,436

Table 3. Concentration indices and the informal-formal gap.

Indices are ranked on per-capita consumption; a negative value means the outcome is concentrated among the poor. Erreygers is the bounded correction for binary outcomes. The sector gap is tested with a design-based Wald test; the bottom panel reports the informal coefficient from a population-weighted linear probability model with standard errors clustered on the enumeration area.

Outcome	Mean	CI	95% low	95% high	Erreygers	SE
CHE, budget share > 10%	0.141	0.040	-0.018	0.098	0.023	0.017
CHE, budget share > 25%	0.034	0.065	-0.088	0.218	0.009	0.010
CHE, capacity to pay >= 40%	0.079	-0.150	-0.223	-0.077	-0.047	0.013
OOP share of consumption	0.048	0.032	-0.013	0.077		
Sought care when ill (any member)	0.689	-0.009	-0.029	0.010	-0.025	0.027

Outcome (%)	Informal	SE	Formal	SE	Difference pp	p
CHE > 10%	14.4	0.9	13.3	1.7	1.1	0.591
CHE > 25%	3.6	0.5	2.5	0.9	1.1	0.303
CHE, CTP >= 40%	8.7	0.7	4.8	1.2	3.8	0.005
OOP share of consumption	4.8	0.2	4.6	0.4	0.3	0.536

Outcome	Controls	Informal coefficient pp	SE	p
CHE > 10%	Unadjusted	1.1	1.9	0.571
CHE > 10%	Consumption quintile	1.6	1.8	0.401
CHE > 10%	Quintile, zone, residence, household size	1.3	1.9	0.486
CHE, CTP >= 40%	Unadjusted	3.8	1.4	0.007
CHE, CTP >= 40%	Consumption quintile	3.1	1.4	0.026
CHE, CTP >= 40%	Quintile, zone, residence, household size	2.4	1.5	0.107

Table 4. From observed spending to the gross premium, and what households can pay.

Naira per person per year. Second panel: loadings for a 20,000-life pool on the two risk-margin conventions (other pool sizes in Table A5). Third panel: the gross premium against per-capita consumption by quintile; quintiles rank people, and household size is the household-weighted mean since a premium is billed once per household. Fourth panel: the contribution ceiling, 5% of the mean per-capita consumption of informal-sector members of each quintile.

Step	Naira	Change
Observed out-of-pocket cost per person-year	19,675	
less traditional and spiritual care	19,320	-354
less services outside the basic package (outpatient 85% / inpatient 90% covered)	16,487	-2,833
plus induced demand (arc elasticity -0.20)	23,296	6,808
less member co-payment on drugs (10%)	21,459	-1,837

Risk-margin basis	Pure premium	Risk margin	Admin 15%	Selection 10%	Gross premium
Standard deviation	21,459	81	3,231	2,477	27,247
CVaR 95%	21,459	1,812	3,491	2,676	29,437

Quintile	Consumption per capita	Household size	Premium as % of consumption	Premium per household
Q1 (poorest)	119,503	8.2	22.8	222,993
Q2	203,279	7.6	13.4	206,162
Q3	280,983	6.0	9.7	164,381
Q4	412,333	4.7	6.6	129,121
Q5 (richest)	895,288	3.0	3.0	82,823

Informal-sector group	Consumption per capita	Contribution ceiling (5%)
All informal households	359,726	17,986
Informal, quintile 1	118,923	5,946
Informal, quintile 2	202,745	10,137
Informal, quintile 3	279,669	13,983
Informal, quintile 4	410,435	20,522
Informal, quintile 5	888,378	44,419

Table 5. Minimum subsidy per enrollee for a ruin probability below the target over three years, no opening capital.

Upper panel: by pool size and take-up pattern, at the flat contribution. Lower panel: a 20,000-life pool by take-up rate, where enrollment probability rises with predicted cost by the tilt factor per standard deviation of log predicted cost; at 100% take-up everyone enrolls and no selection is possible.

Pool size	Take-up	Ruin target	Subsidy	% of gross premium	N bn per 100,000 enrollees
5,000	Moderate adverse selection	1%	36,637	134	3.66
5,000	Moderate adverse selection	5%	34,581	127	3.46
5,000	Random	1%	18,553	68	1.86
5,000	Random	5%	17,083	63	1.71
5,000	Strong adverse selection	1%	57,024	209	5.70
5,000	Strong adverse selection	5%	54,360	200	5.44
20,000	Moderate adverse selection	1%	33,122	122	3.31
20,000	Moderate adverse selection	5%	32,205	118	3.22
20,000	Random	1%	16,019	59	1.60
20,000	Random	5%	15,339	56	1.53
20,000	Strong adverse selection	1%	52,559	193	5.26

Pool size	Take-up	Ruin target	Subsidy	% of gross premium	N bn per 100,000 enrollees
20,000	Strong adverse selection	5%	51,401	189	5.14
100,000	Moderate adverse selection	1%	31,371	115	3.14
100,000	Moderate adverse selection	5%	30,945	114	3.09
100,000	Random	1%	14,700	54	1.47
100,000	Random	5%	14,408	53	1.44
100,000	Strong adverse selection	1%	50,300	185	5.03
100,000	Strong adverse selection	5%	49,819	183	4.98

Tilt	Take-up	Expected claim	Selection loading %	Subsidy, ruin < 5%	% of gross premium
Moderate tilt	10%	35,951	68	32,233	118
Moderate tilt	25%	35,938	68	32,308	119
Moderate tilt	50%	31,904	49	27,612	101
Moderate tilt	75%	26,198	23	20,992	77
Moderate tilt	100%	21,342	0	15,358	56
Strong tilt	10%	52,716	147	51,486	189
Strong tilt	25%	45,208	112	42,790	157
Strong tilt	50%	34,168	60	30,197	111
Strong tilt	75%	26,540	24	21,351	78
Strong tilt	100%	21,342	0	15,358	56

Table 6. Financing designs for a 20,000-life pool: contribution schedules and per-life excess-of-loss reinsurance.

Minimum subsidy holds the three-year ruin probability below 5%. Reinsurance cedes each person's annual cost above the retention for a premium of 1.3 times the expected ceded cost.

Contribution schedule	Take-up	Mean contribution	Expected claim	Minimum subsidy	Subsidy share
Flat (Q1-Q3 ceiling for all)	Random	10,022	21,342	15,339	60%
Flat (Q1-Q3 ceiling for all)	Strong adverse selection	10,022	52,718	51,401	84%
Graded (each quintile's own ceiling)	Random	17,986	21,342	7,375	29%

Contribution schedule	Take-up	Mean contribution	Expected claim	Minimum subsidy	Subsidy share
Graded (each quintile's own ceiling)	Strong adverse selection	26,055	52,718	35,369	58%
Q1-Q2 exempt, Q3-Q5 graded	Random	14,529	21,342	10,831	43%
Q1-Q2 exempt, Q3-Q5 graded	Strong adverse selection	24,108	52,718	37,315	61%

Take-up	Retention	Reinsurance premium	SD of claims per enrollee	p99 of claims per enrollee	Minimum subsidy
Random	None	0	850	23,426	15,334
Random	250,000	8,260	315	23,950	16,399
Random	500,000	4,812	427	23,468	15,726
Random	1,000,000	2,654	554	23,240	15,403
Strong adverse selection	None	0	1,477	56,201	51,388
Strong adverse selection	250,000	28,937	475	60,494	56,853
Strong adverse selection	500,000	17,322	717	58,369	54,312
Strong adverse selection	1,000,000	8,671	981	57,091	52,585

Table 7. Change in catastrophic expenditure under coverage scenarios.

Paired design-based test of the change on the same households. A negative reduction means catastrophic expenditure rises. Levels for every scenario and quintile are in Tables A9 and A10.

Scenario	Payment basis	Measure	Baseline %	Counterfactual %	Reduction pp	SE	p
Vulnerable group only, fully subsidised	Out-of-pocket only	Budget share > 10%	14.1	10.4	3.8	0.5	0.000
Vulnerable group only, fully subsidised	Out-of-pocket only	Capacity to pay >= 40%	7.9	4.5	3.4	0.5	0.000
Vulnerable group only,	OOP plus member	Budget share > 10%	14.1	10.4	3.8	0.5	0.000

Scenario	Payment basis	Measure	Baseline %	Counterfactual %	Reduction pp	SE	p
fully subsidised	contribution						
Vulnerable group only, fully subsidised	OOP plus member contribution	Capacity to pay >= 40%	7.9	4.5	3.4	0.5	0.000
Informal sector, 50% take-up	Out-of-pocket only	Budget share > 10%	14.1	9.3	4.8	0.5	0.000
Informal sector, 50% take-up	Out-of-pocket only	Capacity to pay >= 40%	7.9	4.4	3.5	0.4	0.000
Informal sector, 50% take-up	OOP plus member contribution	Budget share > 10%	14.1	13.2	0.9	0.5	0.086
Informal sector, 50% take-up	OOP plus member contribution	Capacity to pay >= 40%	7.9	8.1	-0.2	0.5	0.731
Informal sector, full coverage	Out-of-pocket only	Budget share > 10%	14.1	4.1	10.0	0.7	0.000
Informal sector, full coverage	Out-of-pocket only	Capacity to pay >= 40%	7.9	1.5	6.4	0.6	0.000
Informal sector, full coverage	OOP plus member contribution	Budget share > 10%	14.1	11.5	2.6	0.7	0.000
Informal sector, full coverage	OOP plus member contribution	Capacity to pay >= 40%	7.9	8.4	-0.5	0.7	0.468
Informal sector, full coverage, no contribution	Out-of-pocket only	Budget share > 10%	14.1	4.4	9.7	0.7	0.000
Informal sector, full coverage, no contribution	Out-of-pocket only	Capacity to pay >= 40%	7.9	1.7	6.2	0.6	0.000
Informal sector, full coverage, no contribution	OOP plus member contribution	Budget share > 10%	14.1	4.4	9.7	0.7	0.000
Informal sector, full coverage,	OOP plus member contribution	Capacity to pay >= 40%	7.9	1.7	6.2	0.6	0.000

Scenario	Payment basis	Measure	Baseline %	Counterfactual %	Reduction pp	SE	p
no contribution							
Informal sector, full coverage, graded contribution	Out-of-pocket only	Budget share > 10%	14.1	4.2	9.9	0.7	0.000
Informal sector, full coverage, graded contribution	Out-of-pocket only	Capacity to pay >= 40%	7.9	1.4	6.5	0.6	0.000
Informal sector, full coverage, graded contribution	OOP plus member contribution	Budget share > 10%	14.1	10.0	4.1	0.5	0.000
Informal sector, full coverage, graded contribution	OOP plus member contribution	Capacity to pay >= 40%	7.9	5.7	2.2	0.5	0.000
Informal sector, full coverage, Q1-Q2 exempt	Out-of-pocket only	Budget share > 10%	14.1	4.3	9.8	0.7	0.000
Informal sector, full coverage, Q1-Q2 exempt	Out-of-pocket only	Capacity to pay >= 40%	7.9	1.6	6.3	0.6	0.000
Informal sector, full coverage, Q1-Q2 exempt	OOP plus member contribution	Budget share > 10%	14.1	7.6	6.6	0.6	0.000
Informal sector, full coverage, Q1-Q2 exempt	OOP plus member contribution	Capacity to pay >= 40%	7.9	2.5	5.4	0.5	0.000
Universal coverage	Out-of-pocket only	Budget share > 10%	14.1	1.6	12.5	0.8	0.000
Universal coverage	Out-of-pocket only	Capacity to pay >= 40%	7.9	0.5	7.4	0.6	0.000
Universal coverage	OOP plus member contribution	Budget share > 10%	14.1	10.0	4.1	0.8	0.000

Scenario	Payment basis	Measure	Baseline %	Counterfactual %	Reduction pp	SE	p
Universal coverage	OOP plus member contribution	Capacity to pay >= 40%	7.9	7.9	-0.0	0.9	0.979

Table 8. Recall-treatment bounds and design-based bootstrap intervals.

Upper panel: the 4-week outpatient window scaled to a year (full within-year persistence) against annual cost drawn as independent 4-week windows from the fitted frequency model (no persistence), and two smaller annualizers. Subsidies are for a 20,000-life pool over three years, ruin below 5%. Lower panel: Rao-Wu rescaled bootstrap over enumeration areas within strata, 200 replicates, percentile intervals.

Recall treatment	CHE10 %	CTP40 %	Pure premium	Gross premium	Subsidy, random	Subsidy, strong selection
x13, window scaled to a year (headline, full persistence)	14.1	7.9	21,459	27,247	15,388	51,439
x13, independent 4-week windows (no persistence)	6.7	3.3	21,793	27,612	13,922	38,091
x12, window scaled	13.3	7.1	19,944	25,324	13,574	47,208
x10, window scaled	11.2	5.9	16,914	21,478	10,076	38,788

Quantity	Point estimate	Bootstrap SE	2.5%	97.5%
CHE, budget share > 10% (%)	14.1	0.9	12.4	15.9
CHE, capacity to pay >= 40% (%)	7.9	0.6	6.8	9.3
Pure premium (N)	21,458.5	1,291.3	19,118.6	23,928.6
Gross premium (N)	27,247.1	1,643.9	24,270.3	30,387.5
Flat contribution (N)	10,022.3	44.0	9,925.0	10,100.0
Expected claim, random take-up (N)	21,342.0	1,484.6	18,769.5	24,183.6

Quantity	Point estimate	Bootstrap SE	2.5%	97.5%
Minimum subsidy, random take-up (N)	15,343.9	1,849.7	12,049.8	18,947.8
Expected claim, strong selection (N)	52,717.9	4,808.2	43,928.2	62,124.5
Minimum subsidy, strong selection (N)	51,406.7	5,606.5	41,075.8	62,337.2

Appendix tables

Table A1. The rebuilt consumption aggregate against wave 4's published aggregate, and the calibration check.

Upper panel: component means. Lower panel: CHE on wave 4 computed with the official aggregate, the rebuilt aggregate and the calibrated rebuilt aggregate, holding out-of-pocket spending fixed. Calibration factors: food 1.158, non-food 0.892, rent to non-rent 0.0687.

Component	Official mean	Rebuilt mean	Ratio	Pearson r	Spearman rho
Food	544,235	470,092	0.864	0.428	0.742
Non-food incl. health	230,380	206,996	0.898	0.722	0.775
Total excl. rent & education	774,615	677,087	0.874	0.576	0.798
Quintile agreement (exact / within one)			0.541	0.920	

Wave-4 aggregate	Mean	Ratio to official	Spearman	CHE10 %	CHE25 %	Mean OOP share %
Official aggregate	1,041,090	1.000	1.000	14.1	2.5	4.9
Rebuilt, uncalibrated	873,100	0.839	0.850	17.6	4.0	5.9
Rebuilt, calibrated	1,000,751	0.961	0.841	15.2	2.7	5.2

Table A2. Catastrophic expenditure by residence and zone, and observed health-insurance coverage.

Measure	Dimension	Group	Incidence %	95% low	95% high	n
Budget share > 10%	Residence	Rural	14.2	12.0	16.4	3,205

Measure	Dimension	Group	Incidence %	95% low	95% high	n
Budget share > 10%	Residence	Urban	14.0	11.4	16.6	1,480
Budget share > 10%	Zone	North Central	21.3	16.4	26.1	804
Budget share > 10%	Zone	North East	13.5	10.4	16.7	808
Budget share > 10%	Zone	North West	9.2	6.0	12.5	773
Budget share > 10%	Zone	South East	24.8	20.2	29.5	786
Budget share > 10%	Zone	South South	15.9	11.2	20.5	762
Budget share > 10%	Zone	South West	8.8	6.1	11.4	752
Budget share > 25%	Residence	Rural	3.8	2.6	4.9	3,205
Budget share > 25%	Residence	Urban	2.6	1.4	3.8	1,480
Budget share > 25%	Zone	North Central	5.2	2.2	8.2	804
Budget share > 25%	Zone	North East	2.3	0.6	4.0	808
Budget share > 25%	Zone	North West	1.8	0.1	3.5	773
Budget share > 25%	Zone	South East	7.3	4.9	9.7	786
Budget share > 25%	Zone	South South	4.5	2.0	6.9	762
Budget share > 25%	Zone	South West	1.9	0.9	3.0	752

Measure	Dimension	Group	Incidence %	95% low	95% high	n
Budget share > 40%	Residence	Rural	1.0	0.5	1.4	3,205
Budget share > 40%	Residence	Urban	0.5	0.1	0.9	1,480
Budget share > 40%	Zone	North Central	0.9	-0.3	2.1	804
Budget share > 40%	Zone	North East	1.2	0.2	2.1	808
Budget share > 40%	Zone	North West	0.2	-0.2	0.5	773
Budget share > 40%	Zone	South East	1.6	0.5	2.7	786
Budget share > 40%	Zone	South South	1.0	0.3	1.7	762
Budget share > 40%	Zone	South West	0.7	0.0	1.4	752
Capacity to pay >= 40%	Residence	Rural	9.0	7.4	10.7	3,205
Capacity to pay >= 40%	Residence	Urban	6.0	4.2	7.8	1,480
Capacity to pay >= 40%	Zone	North Central	13.0	9.1	17.0	804
Capacity to pay >= 40%	Zone	North East	10.9	8.2	13.6	808
Capacity to pay >= 40%	Zone	North West	4.6	2.2	6.9	773
Capacity to pay >= 40%	Zone	South East	13.7	9.9	17.5	786
Capacity to pay >= 40%	Zone	South South	7.3	4.5	10.1	762

Measure	Dimension	Group	Incidence %	95% low	95% high	n
Capacity to pay >= 40%	Zone	South West	3.5	1.9	5.1	752

Dimension	Group	Households covered %	SE	Individuals covered %	Households paying a premium %	n
Overall	All households	1.98	0.34	1.07	0.20	4,685
Sector	Informal	0.57	0.20	0.33	0.03	3,876
Sector	Formal	8.03	1.44	4.02	0.94	809
Consumption quintile	Q1 (poorest)	0.15	0.09	0.08	0.00	701
Consumption quintile	Q2	1.45	0.80	0.50	0.06	699
Consumption quintile	Q3	1.10	0.38	0.60	0.20	808
Consumption quintile	Q4	2.21	0.76	1.46	0.08	1,041
Consumption quintile	Q5 (richest)	3.15	0.68	2.71	0.42	1,436
Residence	Urban	3.05	0.66	1.81	0.25	1,480
Residence	Rural	1.29	0.36	0.64	0.17	3,205
Zone	North Central	2.75	1.22	1.72	0.17	804
Zone	North East	1.31	0.44	0.65	0.04	808
Zone	North West	1.27	0.55	0.56	0.00	773
Zone	South East	1.93	0.73	0.99	0.94	786
Zone	South South	1.91	0.70	1.27	0.08	762
Zone	South West	2.40	0.85	1.50	0.10	752

Table A3. Impoverishment from out-of-pocket payments, and its sensitivity to the poverty line.

The headcount is a level and depends on the line and on the consumption aggregate; the impoverishing effect is a difference and moves much less.

Indicator	Estimate	SE
Poverty line (N per person per year, wave-5 prices)	317,463.30	nan
Poverty headcount before OOP (%)	57.20	1.65
Poverty headcount after OOP (%)	60.29	1.60
Impoverished by OOP (pp)	3.09	nan

Indicator	Estimate	SE
Newly impoverished households (%)	3.09	0.46
Normalized poverty gap before OOP (%)	22.01	0.93
Normalized poverty gap after OOP (%)	23.87	0.93
Increase in poverty gap (pp)	1.86	nan

Poverty line	x national line	Headcount before %	Headcount after %	Impoverishment pp	Millions
158,732	0.50	17.6	19.9	2.30	4.87
238,097	0.75	40.2	42.8	2.67	5.64
317,463	1.00	57.2	60.3	3.09	6.54
396,829	1.25	69.9	71.5	1.63	3.45
476,195	1.50	76.1	78.2	2.14	4.52

Table A4. Decomposition of the concentration indices.

Descriptive, not causal. Contributions are in index units; shares are not shown for the budget-share index because it is not distinguishable from zero and shares of a near-zero total are not interpretable.

Outcome	Variable	Beta	Mean x	CI of x	Contribution
CHE, budget share > 10%	Rural	0.0063	0.636	-0.0909	-0.0026
CHE, budget share > 10%	Household size	-0.0061	7.401	-0.1691	0.0539
CHE, budget share > 10%	Female head	-0.0074	0.131	0.1761	-0.0012
CHE, budget share > 10%	Head aged 60+	-0.0552	0.276	0.0247	-0.0027
CHE, budget share > 10%	Any child under 5	0.0076	0.355	-0.1972	-0.0038
CHE, budget share > 10%	Any member 60+	0.0731	0.335	-0.0228	-0.0040
CHE, budget share > 10%	Severe functional difficulty	0.2043	0.098	-0.0013	-0.0002
CHE, budget share > 10%	Informal sector	0.0079	0.800	-0.0497	-0.0022
CHE, budget share > 10%	Hospitalization in past year	0.3619	0.130	0.0782	0.0261
CHE, budget share > 10%	Residual				-0.0234
CHE, capacity to pay >=	Rural	0.0257	0.636	-0.0909	-0.0188

Outcome	Variable	Beta	Mean x	CI of x	Contribution
40%					
CHE, capacity to pay >= 40%	Household size	-0.0020	7.401	-0.1691	0.0322
CHE, capacity to pay >= 40%	Female head	-0.0068	0.131	0.1761	-0.0020
CHE, capacity to pay >= 40%	Head aged 60+	-0.0483	0.276	0.0247	-0.0042
CHE, capacity to pay >= 40%	Any child under 5	0.0054	0.355	-0.1972	-0.0048
CHE, capacity to pay >= 40%	Any member 60+	0.0629	0.335	-0.0228	-0.0061
CHE, capacity to pay >= 40%	Severe functional difficulty	0.1677	0.098	-0.0013	-0.0003
CHE, capacity to pay >= 40%	Informal sector	0.0288	0.800	-0.0497	-0.0145
CHE, capacity to pay >= 40%	Hospitalization in past year	0.1900	0.130	0.0782	0.0245
CHE, capacity to pay >= 40%	Residual				-0.1562

Table A5. Gross premium by pool size, risk margins, risk relativities and the comparison with a published state-scheme rate.

Relativities are relative to the community average; the premium itself is community-rated. The Lagos comparison is a plausibility check: the Ilera Eko package is not the NHIA basic package, and its rates are nominal July-2024 naira.

Pool size	Risk-margin basis	Risk margin	Gross premium
5,000	Standard deviation	161	27,349
5,000	CVaR 95%	3,528	31,608
20,000	Standard deviation	81	27,247
20,000	CVaR 95%	1,812	29,437
100,000	Standard deviation	36	27,191
100,000	CVaR 95%	748	28,092

Pool size	SD principle	CVaR 95%	SD of the pool mean
5,000	161	3,528	1,578
20,000	81	1,812	820
100,000	36	748	355

Dimension	Class	n	Mean insurer cost	Relativity
Age band	0-4	1,674	10,898	0.51
Age band	15-24	5,268	9,400	0.44
Age band	25-44	5,326	28,985	1.35
Age band	45-59	2,810	34,581	1.61
Age band	5-14	7,492	9,144	0.43
Age band	60+	2,059	71,110	3.31
Sex	Male	12,414	21,661	1.01
Sex	Female	12,215	21,244	0.99
Residence	Rural	17,413	20,551	0.96
Residence	Urban	7,216	23,045	1.07
Consumption quintile	1	5,642	5,618	0.26
Consumption quintile	2	5,062	10,337	0.48
Consumption quintile	3	4,874	13,540	0.63
Consumption quintile	4	4,886	22,696	1.06
Consumption quintile	5	4,165	55,058	2.57
Severe functional difficulty	No	24,071	18,966	0.88
Severe functional difficulty	Yes	558	138,299	6.44
Sector	Formal	4,588	21,926	1.02
Sector	Informal	20,041	21,342	0.99

Scheme	Published premium	Modeled premium	Ratio
Lagos LASHMA 'Ilera Eko' Standard, individual	15,000	27,247	1.82
Lagos LASHMA 'Ilera Eko' Standard, family of four	55,000	27,247	0.50
Lagos LASHMA 'Ilera Eko' Standard, family of six	80,000	27,247	0.34

Table A6. Frequency, severity and Tweedie model estimates.

Survey-weighted, standard errors clustered on the enumeration area. exp(coef) is the multiplicative effect on the fitted mean.

Model	Term	Coef	SE	z	p	exp(coef)
Frequency: outpatient (Poisson, annual rate)	Intercept	0.6643	0.1571	4.23	0.000	1.943
Frequency: outpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.0-4]	0.1857	0.0824	2.25	0.024	1.204
Frequency: outpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.15-24]	-0.4115	0.0600	-6.86	0.000	0.663
Frequency: outpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.45-59]	0.1615	0.0497	3.25	0.001	1.175
Frequency: outpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.5-14]	-0.1729	0.0534	-3.24	0.001	0.841
Frequency: outpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.60+]	0.5032	0.0584	8.61	0.000	1.654
Frequency: outpatient (Poisson, annual rate)	C(zone)[T.North East]	0.0974	0.1093	0.89	0.373	1.102
Frequency: outpatient (Poisson, annual rate)	C(zone)[T.North West]	-0.0840	0.1086	-0.77	0.439	0.919
Frequency: outpatient (Poisson, annual rate)	C(zone)[T.South East]	0.0835	0.1104	0.76	0.450	1.087
Frequency: outpatient (Poisson, annual rate)	C(zone)[T.South South]	0.1016	0.1058	0.96	0.337	1.107

Model	Term	Coef	SE	z	p	exp(coef)
Frequency: outpatient (Poisson, annual rate)	C(zone)[T.South West]	-0.3365	0.1099	-3.06	0.002	0.714
Frequency: outpatient (Poisson, annual rate)	C(quintile)[T.2]	0.2667	0.0841	3.17	0.002	1.306
Frequency: outpatient (Poisson, annual rate)	C(quintile)[T.3]	0.4864	0.0803	6.06	0.000	1.627
Frequency: outpatient (Poisson, annual rate)	C(quintile)[T.4]	0.5343	0.0814	6.56	0.000	1.706
Frequency: outpatient (Poisson, annual rate)	C(quintile)[T.5]	0.6872	0.0858	8.01	0.000	1.988
Frequency: outpatient (Poisson, annual rate)	female	0.0693	0.0335	2.07	0.038	1.072
Frequency: outpatient (Poisson, annual rate)	urban	0.0476	0.0595	0.80	0.424	1.049
Frequency: outpatient (Poisson, annual rate)	chronic	0.5330	0.0603	8.84	0.000	1.704
Frequency: outpatient (Poisson, annual rate)	hhsiz	-0.0119	0.0068	-1.75	0.081	0.988
Frequency: inpatient (Poisson, annual rate)	Intercept	-4.0191	0.4306	-9.33	0.000	0.018

Model	Term	Coef	SE	z	p	exp(coef)
Frequency: inpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.0-4]	-0.5526	0.2863	-1.93	0.054	0.575
Frequency: inpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.15-24]	-0.4628	0.1596	-2.90	0.004	0.630
Frequency: inpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.45-59]	0.1292	0.1334	0.97	0.333	1.138
Frequency: inpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.5-14]	-0.5592	0.1453	-3.85	0.000	0.572
Frequency: inpatient (Poisson, annual rate)	C(age_band, Treatment(reference='25- 44'))[T.60+]	0.4225	0.1670	2.53	0.011	1.526
Frequency: inpatient (Poisson, annual rate)	C(zone)[T.North East]	-0.7250	0.2421	-2.99	0.003	0.484
Frequency: inpatient (Poisson, annual rate)	C(zone)[T.North West]	-0.9135	0.2410	-3.79	0.000	0.401
Frequency: inpatient (Poisson, annual rate)	C(zone)[T.South East]	-0.9122	0.2032	-4.49	0.000	0.402
Frequency: inpatient (Poisson, annual rate)	C(zone)[T.South South]	-0.9894	0.2342	-4.22	0.000	0.372
Frequency: inpatient (Poisson, annual rate)	C(zone)[T.South West]	-1.7381	0.2534	-6.86	0.000	0.176

Model	Term	Coef	SE	z	p	exp(coef)
Frequency: inpatient (Poisson, annual rate)	C(quintile)[T.2]	0.9629	0.3213	3.00	0.003	2.619
Frequency: inpatient (Poisson, annual rate)	C(quintile)[T.3]	0.9662	0.3396	2.84	0.004	2.628
Frequency: inpatient (Poisson, annual rate)	C(quintile)[T.4]	1.2518	0.3425	3.66	0.000	3.497
Frequency: inpatient (Poisson, annual rate)	C(quintile)[T.5]	1.7461	0.3548	4.92	0.000	5.732
Frequency: inpatient (Poisson, annual rate)	female	0.0985	0.0940	1.05	0.295	1.103
Frequency: inpatient (Poisson, annual rate)	urban	0.4503	0.1644	2.74	0.006	1.569
Frequency: inpatient (Poisson, annual rate)	chronic	1.0763	0.1853	5.81	0.000	2.934
Frequency: inpatient (Poisson, annual rate)	hhsz	0.0161	0.0245	0.65	0.513	1.016
Severity: cost per outpatient episode (gamma)	Intercept	8.3841	0.1854	45.21	0.000	4,376.717
Severity: cost per outpatient episode (gamma)	C(age_band, Treatment(reference='25- 44'))[T.0-4]	-0.7646	0.1202	-6.36	0.000	0.465

Model	Term	Coef	SE	z	p	exp(coef)
Severity: cost per outpatient episode (gamma)	C(age_band, Treatment(reference='25- 44'))[T.15-24]	-0.5272	0.1164	-4.53	0.000	0.590
Severity: cost per outpatient episode (gamma)	C(age_band, Treatment(reference='25- 44'))[T.45-59]	-0.0938	0.1347	-0.70	0.486	0.910
Severity: cost per outpatient episode (gamma)	C(age_band, Treatment(reference='25- 44'))[T.5-14]	-0.6960	0.1115	-6.24	0.000	0.499
Severity: cost per outpatient episode (gamma)	C(age_band, Treatment(reference='25- 44'))[T.60+]	0.0582	0.1337	0.44	0.663	1.060
Severity: cost per outpatient episode (gamma)	C(zone)[T.North East]	-0.2288	0.1139	-2.01	0.045	0.796
Severity: cost per outpatient episode (gamma)	C(zone)[T.North West]	-0.4364	0.1121	-3.89	0.000	0.646
Severity: cost per outpatient episode (gamma)	C(zone)[T.South East]	-0.0036	0.1230	-0.03	0.976	0.996
Severity: cost per outpatient episode (gamma)	C(zone)[T.South South]	-0.0826	0.1170	-0.71	0.480	0.921
Severity: cost per outpatient episode (gamma)	C(zone)[T.South West]	-0.1729	0.1651	-1.05	0.295	0.841
Severity: cost per outpatient episode (gamma)	C(quintile)[T.2]	0.2127	0.0982	2.17	0.030	1.237

Model	Term	Coef	SE	z	p	exp(coef)
Severity: cost per outpatient episode (gamma)	C(quintile)[T.3]	0.2573	0.0857	3.00	0.003	1.293
Severity: cost per outpatient episode (gamma)	C(quintile)[T.4]	0.5376	0.0895	6.01	0.000	1.712
Severity: cost per outpatient episode (gamma)	C(quintile)[T.5]	1.1133	0.1073	10.38	0.000	3.044
Severity: cost per outpatient episode (gamma)	female	-0.1430	0.0627	-2.28	0.023	0.867
Severity: cost per outpatient episode (gamma)	urban	0.0666	0.0929	0.72	0.473	1.069
Severity: cost per outpatient episode (gamma)	chronic	0.4834	0.1056	4.58	0.000	1.622
Severity: cost per outpatient episode (gamma)	hhsiz	0.0375	0.0082	4.59	0.000	1.038
Severity: cost per inpatient episode (gamma)	Intercept	9.4126	0.3437	27.39	0.000	12,242.037
Severity: cost per inpatient episode (gamma)	C(age_band, Treatment(reference='25- 44'))[T.0-4]	-1.2628	0.2948	-4.28	0.000	0.283
Severity: cost per inpatient episode (gamma)	C(age_band, Treatment(reference='25- 44'))[T.15-24]	-0.8997	0.2430	-3.70	0.000	0.407

Model	Term	Coef	SE	z	p	exp(coef)
Severity: cost per inpatient episode (gamma)	Treatment(reference='25-44'))[T.45-59]	0.4119	0.3211	1.28	0.200	1.510
Severity: cost per inpatient episode (gamma)	Treatment(reference='25-44'))[T.5-14]	-0.3087	0.3230	-0.96	0.339	0.734
Severity: cost per inpatient episode (gamma)	Treatment(reference='25-44'))[T.60+]	0.2488	0.2733	0.91	0.363	1.282
Severity: cost per inpatient episode (gamma)	C(quintile)[T.2]	1.0855	0.3949	2.75	0.006	2.961
Severity: cost per inpatient episode (gamma)	C(quintile)[T.3]	1.0196	0.3273	3.12	0.002	2.772
Severity: cost per inpatient episode (gamma)	C(quintile)[T.4]	1.6177	0.3880	4.17	0.000	5.042
Severity: cost per inpatient episode (gamma)	C(quintile)[T.5]	1.7803	0.3343	5.33	0.000	5.932
Severity: cost per inpatient episode (gamma)	female	-0.1041	0.1670	-0.62	0.533	0.901
Severity: cost per inpatient episode (gamma)	urban	-0.3804	0.1886	-2.02	0.044	0.684
Aggregate annual cost (Tweedie, p=1.65)	Intercept	9.0016	0.1937	46.48	0.000	8,116.020

Model	Term	Coef	SE	z	p	exp(coef)
Aggregate annual cost (Tweedie, p=1.65)	C(age_band, Treatment(reference='25-44'))[T.0-4]	-0.5734	0.1573	-3.65	0.000	0.564
Aggregate annual cost (Tweedie, p=1.65)	C(age_band, Treatment(reference='25-44'))[T.15-24]	-0.9633	0.1424	-6.77	0.000	0.382
Aggregate annual cost (Tweedie, p=1.65)	C(age_band, Treatment(reference='25-44'))[T.45-59]	0.0914	0.1631	0.56	0.575	1.096
Aggregate annual cost (Tweedie, p=1.65)	C(age_band, Treatment(reference='25-44'))[T.5-14]	-0.8343	0.1327	-6.29	0.000	0.434
Aggregate annual cost (Tweedie, p=1.65)	C(age_band, Treatment(reference='25-44'))[T.60+]	0.5631	0.1639	3.44	0.001	1.756
Aggregate annual cost (Tweedie, p=1.65)	C(zone)[T.North East]	-0.2028	0.1247	-1.63	0.104	0.816
Aggregate annual cost (Tweedie, p=1.65)	C(zone)[T.North West]	-0.5706	0.1207	-4.73	0.000	0.565
Aggregate annual cost (Tweedie, p=1.65)	C(zone)[T.South East]	0.0175	0.1235	0.14	0.887	1.018
Aggregate annual cost (Tweedie, p=1.65)	C(zone)[T.South South]	-0.0538	0.1298	-0.41	0.679	0.948
Aggregate annual cost (Tweedie, p=1.65)	C(zone)[T.South West]	-0.5348	0.1726	-3.10	0.002	0.586

Model	Term	Coef	SE	z	p	exp(coef)
Aggregate annual cost (Tweedie, p=1.65)	C(quintile)[T.2]	0.5689	0.1239	4.59	0.000	1.766
Aggregate annual cost (Tweedie, p=1.65)	C(quintile)[T.3]	0.8525	0.1186	7.19	0.000	2.345
Aggregate annual cost (Tweedie, p=1.65)	C(quintile)[T.4]	1.2091	0.1231	9.82	0.000	3.351
Aggregate annual cost (Tweedie, p=1.65)	C(quintile)[T.5]	1.9367	0.1425	13.59	0.000	6.936
Aggregate annual cost (Tweedie, p=1.65)	female	-0.0539	0.0745	-0.72	0.469	0.948
Aggregate annual cost (Tweedie, p=1.65)	urban	0.0481	0.1104	0.44	0.663	1.049
Aggregate annual cost (Tweedie, p=1.65)	chronic	1.1781	0.1635	7.21	0.000	3.248
Aggregate annual cost (Tweedie, p=1.65)	hhsiz	0.0250	0.0100	2.49	0.013	1.025

Table A7. Cost-model fit statistics, the Tweedie profile likelihood, and calibration by decile of prediction.

Statistic	Value
Outpatient contact rate per person-year (weighted)	2.749
Inpatient episodes per person-year (weighted)	0.030
Mean cost per outpatient episode (N)	6,977.425
Mean cost per inpatient episode (N)	43,985.492

Statistic	Value
Mean annual cost per person (N)	19,674.594
Tweedie variance power p	1.650
Tweedie dispersion phi	428.478
Pearson dispersion, frequency model	0.775
AIC gamma severity	98,131.114
AIC lognormal severity	95,298.669
Quantile-residual KS p-value	0.000

p	Log-likelihood	Deviance
1.20	-92,450.1	228,629,912.3
1.25	-87,297.4	139,618,667.9
1.30	-84,036.2	86,008,598.8
1.35	-81,854.4	53,478,242.2
1.40	-80,350.9	33,586,615.5
1.45	-79,307.6	21,326,453.7
1.50	-78,599.0	13,707,933.5
1.55	-78,152.7	8,934,017.0
1.60	-77,930.7	5,917,429.0
1.65	-77,923.3	3,996,089.7
1.70	-78,151.1	2,764,579.2
1.75	-78,680.8	1,974,035.1
1.80	-79,670.5	1,473,222.3
1.85	-81,512.4	1,176,606.2

Decile	n	Predicted mean	Observed mean	Observed / predicted	Lift
1	2,463	2,894	3,104	1.072	0.16
2	2,463	4,365	4,039	0.925	0.21
3	2,463	5,837	6,121	1.049	0.31
4	2,463	7,535	7,538	1.000	0.38
5	2,463	9,338	10,093	1.081	0.51
6	2,462	12,078	12,675	1.049	0.64
7	2,463	15,781	12,853	0.814	0.65
8	2,463	21,975	18,609	0.847	0.95
9	2,463	32,965	37,389	1.134	1.90
10	2,463	82,286	83,195	1.011	4.23

Table A8. Probability of ruin over three years with no opening capital, and the medical-inflation stress.

The full grid across pool sizes, take-up patterns, subsidy levels, horizons and capital multiples is in output/tables. The stress is a permanent 25% rise in claims from year 2.

Pool size	Take-up	Subsidy	Per enrollee	Ruin probability	MC SE
5,000	Random	0%	0	1.0000	0.0000
5,000	Random	25%	6,812	1.0000	0.0000
5,000	Random	50%	13,624	0.6905	0.0046
5,000	Random	75%	20,435	0.0009	0.0003
5,000	Random	100%	27,247	0.0000	0.0000
5,000	Random	150%	40,871	0.0000	0.0000
5,000	Random	200%	54,494	0.0000	0.0000
20,000	Random	0%	0	1.0000	0.0000
20,000	Random	25%	6,812	1.0000	0.0000
20,000	Random	50%	13,624	0.7195	0.0045
20,000	Random	75%	20,435	0.0000	0.0000
20,000	Random	100%	27,247	0.0000	0.0000
20,000	Random	150%	40,871	0.0000	0.0000
20,000	Random	200%	54,494	0.0000	0.0000
100,000	Random	0%	0	1.0000	0.0000
100,000	Random	25%	6,812	1.0000	0.0000
100,000	Random	50%	13,624	0.7516	0.0043
100,000	Random	75%	20,435	0.0000	0.0000
100,000	Random	100%	27,247	0.0000	0.0000
100,000	Random	150%	40,871	0.0000	0.0000
100,000	Random	200%	54,494	0.0000	0.0000
5,000	Moderate adverse selection	0%	0	1.0000	0.0000
5,000	Moderate adverse selection	25%	6,812	1.0000	0.0000
5,000	Moderate adverse selection	50%	13,624	1.0000	0.0000
5,000	Moderate adverse selection	75%	20,435	1.0000	0.0000
5,000	Moderate adverse selection	100%	27,247	0.9815	0.0013
5,000	Moderate adverse selection	150%	40,871	0.0000	0.0000
5,000	Moderate adverse selection	200%	54,494	0.0000	0.0000

Pool size	Take-up	Subsidy	Per enrollee	Ruin probability	MC SE
20,000	Moderate adverse selection	0%	0	1.0000	0.0000
20,000	Moderate adverse selection	25%	6,812	1.0000	0.0000
20,000	Moderate adverse selection	50%	13,624	1.0000	0.0000
20,000	Moderate adverse selection	75%	20,435	1.0000	0.0000
20,000	Moderate adverse selection	100%	27,247	1.0000	0.0000
20,000	Moderate adverse selection	150%	40,871	0.0000	0.0000
20,000	Moderate adverse selection	200%	54,494	0.0000	0.0000
100,000	Moderate adverse selection	0%	0	1.0000	0.0000
100,000	Moderate adverse selection	25%	6,812	1.0000	0.0000
100,000	Moderate adverse selection	50%	13,624	1.0000	0.0000
100,000	Moderate adverse selection	75%	20,435	1.0000	0.0000
100,000	Moderate adverse selection	100%	27,247	1.0000	0.0000
100,000	Moderate adverse selection	150%	40,871	0.0000	0.0000
100,000	Moderate adverse selection	200%	54,494	0.0000	0.0000
5,000	Strong adverse selection	0%	0	1.0000	0.0000
5,000	Strong adverse selection	25%	6,812	1.0000	0.0000
5,000	Strong adverse selection	50%	13,624	1.0000	0.0000
5,000	Strong adverse selection	75%	20,435	1.0000	0.0000
5,000	Strong adverse selection	100%	27,247	1.0000	0.0000
5,000	Strong adverse selection	150%	40,871	1.0000	0.0000
5,000	Strong adverse selection	200%	54,494	0.0462	0.0021
20,000	Strong adverse selection	0%	0	1.0000	0.0000

Pool size	Take-up	Subsidy	Per enrollee	Ruin probability	MC SE
20,000	Strong adverse selection	25%	6,812	1.0000	0.0000
20,000	Strong adverse selection	50%	13,624	1.0000	0.0000
20,000	Strong adverse selection	75%	20,435	1.0000	0.0000
20,000	Strong adverse selection	100%	27,247	1.0000	0.0000
20,000	Strong adverse selection	150%	40,871	1.0000	0.0000
20,000	Strong adverse selection	200%	54,494	0.0002	0.0001
100,000	Strong adverse selection	0%	0	1.0000	0.0000
100,000	Strong adverse selection	25%	6,812	1.0000	0.0000
100,000	Strong adverse selection	50%	13,624	1.0000	0.0000
100,000	Strong adverse selection	75%	20,435	1.0000	0.0000
100,000	Strong adverse selection	100%	27,247	1.0000	0.0000
100,000	Strong adverse selection	150%	40,871	1.0000	0.0000
100,000	Strong adverse selection	200%	54,494	0.0000	0.0000
Claims shock	Take-up		Minimum subsidy		
0%	Random		15,339		
0%	Moderate adverse selection		32,205		
0%	Strong adverse selection		51,401		
25%	Random		18,723		
25%	Moderate adverse selection		38,067		
25%	Strong adverse selection		60,213		

Table A9. Catastrophic spending under each coverage scenario, on both payment bases.

Scenario	Payment basis	Covered	CHE10 %	CHE25 %	CTP40 %	Poverty after %	Mean payments
Baseline (no coverage)	Out-of-pocket only	0.0%	14.1	3.4	7.9	60.3	108,582

Scenario	Payment basis	Covered	CHE10 %	CHE25 %	CTP40 %	Poverty after %	Mean payments
Baseline (no coverage)	OOP plus member contribution	0.0%	14.1	3.4	7.9	60.3	108,582
Vulnerable group only, fully subsidised	Out-of-pocket only	34.7%	10.4	2.3	4.5	60.3	90,762
Vulnerable group only, fully subsidised	OOP plus member contribution	34.7%	10.4	2.3	4.5	60.3	90,762
Informal sector, 50% take-up	Out-of-pocket only	41.1%	9.3	1.9	4.4	59.7	76,925
Informal sector, 50% take-up	OOP plus member contribution	41.1%	13.2	2.0	8.1	60.3	107,606
Informal sector, full coverage	Out-of-pocket only	80.0%	4.1	0.8	1.5	59.1	46,257
Informal sector, full coverage	OOP plus member contribution	80.0%	11.5	1.2	8.4	60.3	105,844
Informal sector, full coverage, no contribution	Out-of-pocket only	80.0%	4.4	0.8	1.7	60.3	46,257
Informal sector, full coverage, no contribution	OOP plus member contribution	80.0%	4.4	0.8	1.7	60.3	46,257
Informal sector, full coverage, graded contribution	Out-of-pocket only	80.0%	4.2	0.8	1.4	58.3	46,257
Informal sector, full coverage, graded contribution	OOP plus member contribution	80.0%	10.0	1.0	5.7	60.3	132,250
Informal sector, full coverage, Q1-Q2 exempt	Out-of-pocket only	80.0%	4.3	0.8	1.6	58.3	46,257

Scenario	Payment basis	Covered	CHE10 %	CHE25 %	CTP40 %	Poverty after %	Mean payments
Informal sector, full coverage, Q1-Q2 exempt	OOP plus member contribution	80.0%	7.6	0.8	2.5	60.3	105,634
Universal coverage	Out-of-pocket only	100.0%	1.6	0.3	0.5	58.7	27,603
Universal coverage	OOP plus member contribution	100.0%	10.0	0.7	7.9	60.3	101,782

Table A10. Counterfactual catastrophic spending by consumption quintile, out-of-pocket plus contribution basis.

Scenario	Measure	Quintile	Estimate %	95% low	95% high
Baseline (no coverage)	CHE > 10%	Q1 (poorest)	11.9	8.6	15.2
Baseline (no coverage)	CHE > 10%	Q2	14.4	10.3	18.5
Baseline (no coverage)	CHE > 10%	Q3	14.2	10.7	17.7
Baseline (no coverage)	CHE > 10%	Q4	14.6	11.0	18.3
Baseline (no coverage)	CHE > 10%	Q5 (richest)	15.6	12.6	18.6
Baseline (no coverage)	CHE, CTP >= 40%	Q1 (poorest)	9.9	7.3	12.6
Baseline (no coverage)	CHE, CTP >= 40%	Q2	11.2	7.4	15.0
Baseline (no coverage)	CHE, CTP >= 40%	Q3	6.4	4.3	8.4
Baseline (no coverage)	CHE, CTP >= 40%	Q4	7.6	5.1	10.1
Baseline (no coverage)	CHE, CTP >= 40%	Q5 (richest)	4.4	2.8	6.1
Vulnerable group only, fully subsidised	CHE > 10%	Q1 (poorest)	3.5	1.7	5.3
Vulnerable group only, fully subsidised	CHE > 10%	Q2	4.0	1.9	6.1
Vulnerable group only,	CHE > 10%	Q3	14.2	10.7	17.7

Scenario	Measure	Quintile	Estimate %	95% low	95% high
fully subsidised					
Vulnerable group only, fully subsidised	CHE > 10%	Q4	14.6	11.0	18.3
Vulnerable group only, fully subsidised	CHE > 10%	Q5 (richest)	15.6	12.6	18.6
Vulnerable group only, fully subsidised	CHE, CTP >= 40%	Q1 (poorest)	1.6	0.5	2.6
Vulnerable group only, fully subsidised	CHE, CTP >= 40%	Q2	2.4	0.9	3.9
Vulnerable group only, fully subsidised	CHE, CTP >= 40%	Q3	6.4	4.3	8.4
Vulnerable group only, fully subsidised	CHE, CTP >= 40%	Q4	7.6	5.1	10.1
Vulnerable group only, fully subsidised	CHE, CTP >= 40%	Q5 (richest)	4.4	2.8	6.1
Informal sector, 50% take-up	CHE > 10%	Q1 (poorest)	22.5	18.0	26.9
Informal sector, 50% take-up	CHE > 10%	Q2	11.8	8.0	15.6
Informal sector, 50% take-up	CHE > 10%	Q3	9.7	6.7	12.6
Informal sector, 50% take-up	CHE > 10%	Q4	10.4	7.6	13.2
Informal sector, 50% take-up	CHE > 10%	Q5 (richest)	11.7	9.1	14.2
Informal sector, 50% take-up	CHE, CTP >= 40%	Q1 (poorest)	19.2	15.4	23.1

Scenario	Measure	Quintile	Estimate %	95% low	95% high
Informal sector, 50% take-up	CHE, CTP >= 40%	Q2	9.8	6.4	13.3
Informal sector, 50% take-up	CHE, CTP >= 40%	Q3	4.2	2.6	5.9
Informal sector, 50% take-up	CHE, CTP >= 40%	Q4	4.6	2.6	6.6
Informal sector, 50% take-up	CHE, CTP >= 40%	Q5 (richest)	2.4	1.5	3.4
Informal sector, full coverage	CHE > 10%	Q1 (poorest)	31.0	25.6	36.4
Informal sector, full coverage	CHE > 10%	Q2	7.1	4.2	10.0
Informal sector, full coverage	CHE > 10%	Q3	6.0	3.7	8.3
Informal sector, full coverage	CHE > 10%	Q4	7.1	4.6	9.7
Informal sector, full coverage	CHE > 10%	Q5 (richest)	6.5	4.6	8.4
Informal sector, full coverage	CHE, CTP >= 40%	Q1 (poorest)	28.3	23.1	33.6
Informal sector, full coverage	CHE, CTP >= 40%	Q2	7.5	4.5	10.5
Informal sector, full coverage	CHE, CTP >= 40%	Q3	2.4	1.0	3.8
Informal sector, full coverage	CHE, CTP >= 40%	Q4	2.5	0.8	4.1
Informal sector, full coverage	CHE, CTP >= 40%	Q5 (richest)	1.5	0.7	2.3
Informal sector, full coverage, no contribution	CHE > 10%	Q1 (poorest)	3.5	1.7	5.3
Informal sector, full	CHE > 10%	Q2	4.0	1.9	6.1

Scenario	Measure	Quintile	Estimate %	95% low	95% high
coverage, no contribution					
Informal sector, full coverage, no contribution	CHE > 10%	Q3	3.4	1.7	5.1
Informal sector, full coverage, no contribution	CHE > 10%	Q4	5.4	3.2	7.7
Informal sector, full coverage, no contribution	CHE > 10%	Q5 (richest)	5.8	4.0	7.6
Informal sector, full coverage, no contribution	CHE, CTP >= 40%	Q1 (poorest)	1.6	0.5	2.6
Informal sector, full coverage, no contribution	CHE, CTP >= 40%	Q2	2.4	0.9	3.9
Informal sector, full coverage, no contribution	CHE, CTP >= 40%	Q3	1.4	0.3	2.5
Informal sector, full coverage, no contribution	CHE, CTP >= 40%	Q4	2.0	0.4	3.6
Informal sector, full coverage, no contribution	CHE, CTP >= 40%	Q5 (richest)	1.3	0.6	2.1
Informal sector, full coverage, graded contribution	CHE > 10%	Q1 (poorest)	12.6	9.4	15.8
Informal sector, full coverage, graded contribution	CHE > 10%	Q2	7.1	4.2	10.0
Informal sector, full coverage, graded contribution	CHE > 10%	Q3	7.5	4.9	10.0

Scenario	Measure	Quintile	Estimate %	95% low	95% high
Informal sector, full coverage, graded contribution	CHE > 10%	Q4	10.9	7.4	14.4
Informal sector, full coverage, graded contribution	CHE > 10%	Q5 (richest)	12.0	9.3	14.7
Informal sector, full coverage, graded contribution	CHE, CTP >= 40%	Q1 (poorest)	12.3	8.5	16.0
Informal sector, full coverage, graded contribution	CHE, CTP >= 40%	Q2	7.5	4.5	10.5
Informal sector, full coverage, graded contribution	CHE, CTP >= 40%	Q3	3.4	1.9	4.9
Informal sector, full coverage, graded contribution	CHE, CTP >= 40%	Q4	3.5	1.8	5.3
Informal sector, full coverage, graded contribution	CHE, CTP >= 40%	Q5 (richest)	1.9	1.0	2.7
Informal sector, full coverage, Q1-Q2 exempt	CHE > 10%	Q1 (poorest)	3.5	1.7	5.3
Informal sector, full coverage, Q1-Q2 exempt	CHE > 10%	Q2	4.0	1.9	6.1
Informal sector, full coverage, Q1-Q2 exempt	CHE > 10%	Q3	7.5	4.9	10.0
Informal sector, full	CHE > 10%	Q4	10.9	7.4	14.4

Scenario	Measure	Quintile	Estimate %	95% low	95% high
coverage, Q1-Q2 exempt					
Informal sector, full coverage, Q1-Q2 exempt	CHE > 10%	Q5 (richest)	12.0	9.3	14.7
Informal sector, full coverage, Q1-Q2 exempt	CHE, CTP >= 40%	Q1 (poorest)	1.6	0.5	2.6
Informal sector, full coverage, Q1-Q2 exempt	CHE, CTP >= 40%	Q2	2.4	0.9	3.9
Informal sector, full coverage, Q1-Q2 exempt	CHE, CTP >= 40%	Q3	3.4	1.9	4.9
Informal sector, full coverage, Q1-Q2 exempt	CHE, CTP >= 40%	Q4	3.5	1.8	5.3
Informal sector, full coverage, Q1-Q2 exempt	CHE, CTP >= 40%	Q5 (richest)	1.9	1.0	2.7
Universal coverage	CHE > 10%	Q1 (poorest)	32.3	27.0	37.6
Universal coverage	CHE > 10%	Q2	6.0	3.3	8.7
Universal coverage	CHE > 10%	Q3	3.7	2.1	5.3
Universal coverage	CHE > 10%	Q4	4.2	1.9	6.5
Universal coverage	CHE > 10%	Q5 (richest)	3.9	2.3	5.5
Universal coverage	CHE, CTP >= 40%	Q1 (poorest)	29.0	23.3	34.8
Universal coverage	CHE, CTP >= 40%	Q2	6.7	3.9	9.6
Universal coverage	CHE, CTP >= 40%	Q3	1.5	0.4	2.6
Universal coverage	CHE, CTP >= 40%	Q4	1.0	0.3	1.8
Universal coverage	CHE, CTP >= 40%	Q5 (richest)	1.3	0.5	2.1

Table A11. Robustness grid.

Each row changes one decision. Subsidies are for a 20,000-life pool over three years, ruin below 5%, 4,000 simulations.

Check	CHE10 %	CTP40 %	Pure premium	Gross premium	Contribution	Subsidy, random	Subsidy, strong selection
Main specification	14.1	7.9	21,459	27,247	10,022	15,388	51,439
Uncalibrated consumption aggregate	16.7	11.3	21,459	27,247	8,571	16,839	52,890
Equivalence scale hysize ^{0.5}	14.1	7.9	21,459	27,247	10,022	15,388	51,439
Equivalence scale hysize ^{0.75}	14.1	7.9	21,459	27,247	10,022	15,388	51,439
Equivalence scale hysize ^{1.0}	14.1	7.8	21,459	27,247	10,022	15,388	51,439
Outpatient annualiser x12	13.3	7.1	19,944	25,324	9,996	13,574	47,208
Outpatient annualiser x10	11.2	5.9	16,914	21,478	9,930	10,076	38,788
Outpatient annualiser x6	6.0	2.7	10,854	13,787	9,796	3,043	22,032
OOP including transport	15.3	8.8	22,959	29,149	10,065	17,057	55,010
OOP from the wave-5 consumption module	0.1	0.0	21,459	27,247	9,542	15,868	51,920
Top 1% of costs trimmed	13.6	6.6	17,234	21,850	9,994	9,224	31,420
Formal = public-sector employee only	14.1	7.9	21,459	27,247	10,048	14,996	48,675
Formal = any wage employee	14.1	7.9	21,459	27,247	10,019	15,635	51,763
Sector set by the	14.1	7.9	21,459	27,247	10,039	15,331	49,933

Check	CHE10 %	CTP40 %	Pure premium	Gross premium	Contribution	Subsidy, random	Subsidy, strong selection
household head							
Tweedie p fixed at 1.4	14.1	7.9	21,459	27,247	10,022	15,388	51,217
Tweedie p fixed at 1.5	14.1	7.9	21,459	27,247	10,022	15,388	51,423
Tweedie p fixed at 1.7	14.1	7.9	21,459	27,247	10,022	15,388	51,449
Induced-demand elasticity -0.10	14.1	7.9	18,014	22,874	10,022	11,253	41,525
Induced-demand elasticity -0.20	14.1	7.9	21,459	27,247	10,022	15,388	51,439
Induced-demand elasticity -0.35	14.1	7.9	28,240	35,858	10,022	23,393	70,903
Outpatient covered share 70%	14.1	7.9	17,983	22,835	10,022	11,258	41,692
Outpatient covered share 85%	14.1	7.9	21,459	27,247	10,022	15,388	51,439
Outpatient covered share 100%	14.1	7.9	24,934	31,659	10,022	19,434	61,122
Drug co-payment 0%	14.1	7.9	24,731	31,402	10,022	19,232	60,429
Drug co-payment 10%	14.1	7.9	21,459	27,247	10,022	15,388	51,439
Drug co-payment 20%	14.1	7.9	18,738	23,793	10,022	12,154	43,723

Table A12. Wave 4 (2018/19) on the published aggregate, and the multinomial simulation shortcut checked against direct resampling.

The wave-4 out-of-pocket measure comes from the published aggregate, which annualizes a one-month health recall by about twelve; it is not the wave-5 health-module measure.

Wave	Measure	Estimate %	SE
W4 (2018/19)	CHE > 10% (budget share)	14.1	1.0

Wave	Measure	Estimate %	SE
W4 (2018/19)	CHE > 25% (budget share)	2.5	0.3
W4 (2018/19)	CHE, CTP >= 40%	6.7	0.7

Statistic	Multinomial	Direct resampling	Relative difference
mean	106,551,633.8273	106,998,533.4510	-0.0042
sd	8,756,655.2576	8,380,500.3197	0.0449
p95	121,812,991.1793	121,464,785.7158	0.0029
p99	128,191,863.5854	127,976,142.7164	0.0017